G.C.E. (A/L) Examination - 2019

02 - Chemistry (Old Syllabus)

Distribution of Marks

Paper I	:	1 X 50	=	50
Paper II	:			
Part A	:	100 X 4	=	400
Part B	:	150 X 2	=	300
Part C	:	150 X 2	=	300
Total			=	1000
Paper II - Final Marks			=	100

Common Techniques of Marking Answer Scripts.

It is compulsory to adhere to the following standard method in marking answer scripts and entering marks into the mark sheets.

1. Use a red color ball point pen for marking. (Only Chief/Additional Chief Examiner may use a mauve color pen.)
2. Note down Examiner's Code Number and initials on the front page of each answer script.
3. Write off any numerals written wrong with a clear single line and authenticate the alterations with Examiner's initials.
4. Write down marks of each subsection in a △ and write the final marks of each question as a rational number in a □ with the question number. Use the column assigned for Examiners to write down marks.

Example: **Question No. 03**

(i) ..... √ $\frac{4}{5}$

(ii) ..... √ $\frac{3}{5}$

(iii) ..... √ $\frac{3}{5}$

03 (i) $\frac{4}{5}$ + (ii) $\frac{3}{5}$ + (iii) $\frac{3}{5}$ = $\frac{10}{15}$

MCQ answer scripts: (Template)

1. Marking templets for G.C.E.(A/L) and GIT examination will be provided by the Department of Examinations itself. Marking examiners bear the responsibility of using correctly prepared and certified templates.
2. Then, check the answer scripts carefully. If there are more than one or no answers Marked to a certain question write off the options with a line. Sometimes candidates may have erased an option marked previously and selected another option. In such occasions, if the erasure is not clear write off those options too.
3. Place the template on the answer script correctly. Mark the right answers with a 'V' and the wrong answers with a 'X' against the options column. Write down the number of correct answers inside the cage given under each column. Then, add those numbers and write the number of correct answers in the relevant cage.

structured essay type and assay type answer scripts:

1. Cross off any pages left blank by candidates. Underline wrong or unsuitable answers. Show areas where marks can be offered with check marks.

2. Use the right margin of the overland paper to write down the marks.

3. Write down the marks given for each question against the question number in the relevant cage on the front page in two digits. Selection of questions should be in accordance with the instructions given in the question paper. Mark all answers and transfer the marks to the front page, and write off answers with lower marks if extra questions have been answered against instructions.

4. Add the total carefully and write in the relevant cage on the front page. Turn pages of answer script and add all the marks given for all answers again. Check whether that total tallies with the total marks written on the front page.

Preparation of Mark Sheets.

Except for the subjects with a single question paper, final marks of two papers will not be calculated within the evaluation board this time. Therefore, add separate mark sheets for each of the question paper. Write paper 01 marks in the paper 01 column of the mark sheet and write them in words too. Write paper II Marks in the paper II Column and wright the relevant details. For the subject 51 Art, marks for Papers 01, 02 and 03 should be entered numerically in the mark sheets.

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AL/2019/02/E-I(OLD)

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පැරණි නිර්දේශය/பழைய பாடத்திட்டம்/Old Syllabus

OLD

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2019 අගෝස්තු
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2019 ஓகஸ்ற்
General Certificate of Education (Adv. Level) Examination, August 2019

16.08.2019 / 0830 – 1030

රසායන විද්‍යාව I
இரசாயனவியல் I
Chemistry I

02 E I

පැය දෙකයි
இரண்டு மணித்தியாலம்
Two hours

Instructions:

* *Periodic Table is provided.*
* *This paper consists of* **09** *pages.*
* *Answer* ***all*** *the questions.*
* ***Use of calculators is not allowed.***
* *Write your* ***Index Number*** *in the space provided in the answer sheet.*
* *Follow the instructions given on the back of the answer sheet carefully.*
* *In each of the questions* **1** *to* **50**, *pick one of the alternatives from* (1), (2), (3), (4), (5) *which is* ***correct*** *or* ***most appropriate*** *and* ***mark your response on the answer sheet with a cross*** (×) ***in accordance with the instructions given on the back of the answer sheet.***

Universal gas constant	$R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$
Avogadro constant	$N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$
Planck's constant	$h = 6.626 \times 10^{-34}\ \text{J s}$
Velocity of light	$c = 3 \times 10^{8}\ \text{m s}^{-1}$

1. Consider the following statements I and II.

I. The energy absorbed or released by atoms is quantized.

II. The electron in the hydrogen atom travels in a circular path of fixed energy, without emitting energy.

The two scientists who proposed the theories as given by statements I and II respectively are,

(1) Niels Bohr and Albert Einstein
(2) Max Planck and Niels Bohr
(3) Max Planck and Ernest Rutherford
(4) John Dalton and Niels Bohr
(5) Niels Bohr and Max Planck

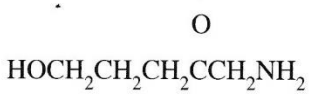
2. The maximum number of **electron pairs** of an atom that are associated with principal quantum number $n = 3$ is,

(1) 3 (2) 4 (3) 5 (4) 8 (5) 9

3. The number of stable resonance structures that can be drawn for the oxalate ion $\left[C_2O_4^{2-} / \left(O_2C-CO_2\right)^{2-}\right]$ is,

(1) 2 (2) 3 (3) 4 (4) 5 (5) 6

4. What is the IUPAC name of the following compound?



(1) 5-hydroxy-2-oxo-1-pentanamine
(2) 1-amino-5-hydroxy-2-oxopentane
(3) 1-amino-5-hydroxy-2-pentanone
(4) 5-hydroxy-1-amino-2-pentanone
(5) 5-amino-4-oxo-1-pentanol

5. Identify the pair of elements which has the **largest** difference in electronegativities.

(1) B and Al (2) Be and Al (3) B and Si (4) B and C (5) Al and C

6. The electron pair geometry and shape around the two nitrogen atoms (labelled as N^1 and N^2) in the H_2NNO molecule (skeleton: $H—\overset{H}{\underset{|}{N^1}}—N^2—O$) respectively are,

N^1		N^2	
(1) tetrahedral	pyramidal	trigonal planar	angular
(2) pyramidal	trigonal planar	trigonal planar	angular
(3) trigonal planar	pyramidal	trigonal planar	trigonal planar
(4) tetrahedral	pyramidal	angular	trigonal planar
(5) tetrahedral	angular	trigonal planar	trigonal planar

7. Which of the following statements is **incorrect** regarding benzene?

(1) The resonance hybrid of benzene is depicted as follows:



(2) All six carbon atoms of benzene are sp^2 hybridized.
(3) The bond lengths between any two carbon atoms of benzene have the same value.
(4) All the C—C—C and the C—C—H bond angles of benzene have the same value.
(5) All the hydrogen atoms of benzene lie in the same plane.

8. $TiCl_4(g)$ reacts with liquid magnesium metal ($Mg(l)$) to give Ti(s) metal and $MgCl_2(l)$ at high temperature. When 0.95 kg of $TiCl_4(g)$ is made to react with 97.2 g of $Mg(l)$, the reactant that is completely consumed (this is commonly referred to as limiting reactant) and the amount of Ti(s) metal formed respectively are, (Molar mass: $TiCl_4 = 190\ g\ mol^{-1}$; $Mg = 24.3\ g\ mol^{-1}$; $Ti = 48\ g\ mol^{-1}$)

(1) $TiCl_4$ and 96 g (2) Mg and 96 g (3) Mg and 48 g
(4) $TiCl_4$ and 192 g (5) Mg and 192 g

9. The ideal gas equation can be expressed in the form, $P = \rho\frac{RT}{M}$ where ρ is the density of the gas, M is the molar mass (g mol^{-1}) of the gas, P is the pressure (Pa) and T is the temperature (K). If the units of R are J mol^{-1} K^{-1}, units of ρ in this equation should be,

(1) kg m^{-3} (2) g m^{-3} (3) g cm^{-3}
(4) g dm^{-3} (5) kg cm^{-3}

10. An ideal binary solution of two volatile liquids **A** and **B** is in equilibrium with its vapour at 25 °C. The mole fractions of **A** in the vapour phase and liquid phase are 0.3 and 0.6 respectively. If the partial pressure of **A** is 30 torr, the total pressure of the system and saturated vapour pressure of **A** respectively are, (1 atm = 760 torr)

(1) 160 torr and 60 torr
(2) 150 torr and 60 torr
(3) 120 torr and 30 torr
(4) 100 torr and 50 torr
(5) 30 torr and 10 torr

11. The correct answer when the chemical species SO_2, SO_3, SO_3^{2-}, SO_4^{2-} and SCl_2 are arranged in the **increasing** order of the electronegativity of sulphur (S) atom is,

(1) $SCl_2 < SO_3^{2-} < SO_2 < SO_3 < SO_4^{2-}$
(2) $SO_3 < SO_4^{2-} < SO_2 < SO_3^{2-} < SCl_2$
(3) $SO_3^{2-} < SO_4^{2-} < SCl_2 < SO_3 < SO_2$
(4) $SCl_2 < SO_3^{2-} < SO_4^{2-} < SO_2 < SO_3$
(5) $SCl_2 < SO_4^{2-} < SO_3^{2-} < SO_2 < SO_3$

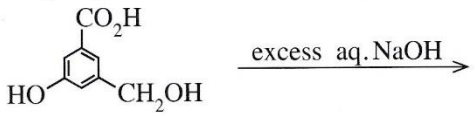
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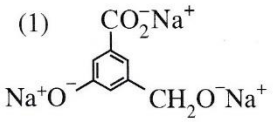
12. Which of the following answers gives the maximum hydroxide concentration that can exist in a 1.775 mol dm^{-3} aqueous solution of $MgCl_2$ at 25 °C? At this temperature, solubility product of $Mg(OH)_2$ is 7.1×10^{-12} mol^3 dm^{-9}.

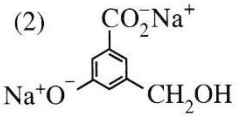
(1) 4.0×10^{-6} mol dm^{-3} (2) 2.0×10^{-6} mol dm^{-3} (3) 1.775×10^{-12} mol dm^{-3}

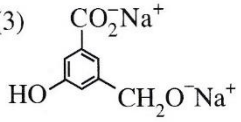
(4) $\sqrt{7.1} \times 10^{-6}$ mol dm^{-3} (5) 1.0×10^{-6} mol dm^{-3}

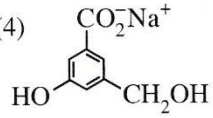
13. What is the major product of the following reaction?

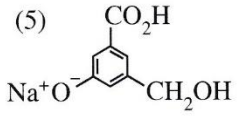


(1) 

(2) 

(3) 

(4) 

(5) 

14. Identify the correct statement from the following.

(1) The bond angle of NF_3 is larger than the bond angle of NH_3.
(2) Elements in group 17 (or 7A) exhibit oxidation states from –1 to +7.
(3) Monoclinic sulphur is the most stable allotrope of sulphur at room temperature.
(4) The density of graphite is higher than the density of diamond.
(5) Aluminium chloride satisfies the octet rule in the gaseous state.

15. The standard electromotive force of the electrochemical cell $Mn(s)\left|Mn^{2+}(aq)\right|\left|Br^-(aq)\right|Br_2(g)\left|Pt(s)\right.$ is 2.27 V. The standard reduction potential of $Br_2(g)\left|Br^-(aq)\right.$ is 1.09 V. The standard reduction potential of $Mn^{2+}(aq)\left|Mn(s)\right.$ is,

(1) –3.36 V (2) –1.18 V (3) 0.59 V (4) 1.18 V (5) 3.36 V

16. The enthalpy change of vaporization and the entropy change of vaporization of a liquid are, 45.00 kJ mol^{-1} and 90.0 J K^{-1} mol^{-1} respectively. The boiling point of the liquid is,

(1) 45.0 °C (2) 62.7 °C (3) 100.0 °C (4) 135.0 °C (5) 227.0 °C

17. What is the **incorrect** statement about $C_6H_5\overset{+}{N}\equiv NCl^-$?

(1) $C_6H_5\overset{+}{N}\equiv NCl^-$ can be obtained by reacting aniline with $HNO_2\left(NaNO_2/HCl\right)$ at 0 – 5 °C.
(2) $C_6H_5\overset{+}{N}\equiv NCl^-$ reacts with KI to give iodobenzene.
(3) The $C_6H_5\overset{+}{N}\equiv N$ ion can act as an electrophile.
(4) When an aqueous solution of $C_6H_5\overset{+}{N}\equiv NCl^-$ is heated, it decomposes to give benzene.
(5) $C_6H_5\overset{+}{N}\equiv NCl^-$ reacts with phenols in a basic medium to give coloured compounds.

18. $H_2S(g)$ reacts with $O_2(g)$ to give only water vapour ($H_2O(g)$) and $SO_2(g)$, as products. When 4 dm^3 of $H_2S(g)$ reacts with 10 dm^3 of $O_2(g)$ at a constant pressure and 250 °C, the final volume of the mixture is,

(1) 6 dm^3 (2) 8 dm^3 (3) 10 dm^3 (4) 12 dm^3 (5) 14 dm^3

[See page four

19. A mixture of A(g) and D(g) was introduced in to a rigid evacuated container at the temperature T. At this temperature, both A(g) and D(g) decompose according to the elementary reactions given below.

$2A(g) \rightarrow B(g) + 3C(g)$; rate constant k_1

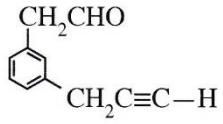
$D(g) \rightarrow B(g) + 2C(g)$

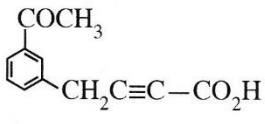
The initial pressure of the container P, changed to 2.7 P after the complete decomposition of both reactants. The initial rate of decomposition of A(g) at this temperature is, (R is the universal gas constant)

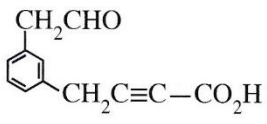
(1) $1.7k_1\left(\frac{P}{RT}\right)$ (2) $2.7k_1\left(\frac{P}{RT}\right)$ (3) $0.09k_1\left(\frac{P}{RT}\right)^2$

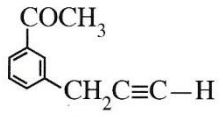
(4) $2.89k_1\left(\frac{P}{RT}\right)^2$ (5) $7.29k_1\left(\frac{P}{RT}\right)^2$

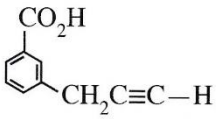
20. An organic compound gives a silver mirror with ammoniacal $AgNO_3$ while it liberates CO_2 with aqueous $NaHCO_3$. The compound could be,

(1) 

(2) 

(3) 

(4) 

(5) 

21. A buffer solution of pH 5.0 was prepared by mixing equal volumes of a 0.10 mol dm^{-3} monobasic weak acid solution and a 0.10 mol dm^{-3} solution of the sodium salt of this acid. The pH of the resultant solution, when 20.00 cm^3 of this buffer solution was mixed with 90.00 cm^3 of 0.10 mol dm^{-3} weak acid solution, is,

(1) 3.0 (2) 4.0 (3) 4.5 (4) 5.5 (5) 6.0

22. Consider the following three aqueous solutions.

P - a weak acid,

Q - an equimolar mixture of the weak acid and its sodium salt,

R - titration mixture at the equivalence point of the titration of the weak acid and a strong base

When each solution is diluted by the same amount at constant temperature, the pH of **P**, **Q** and **R** respectively, will

(1) decrease, increase, not change. (2) increase, not change, decrease.
(3) increase, not change, not change. (4) increase, not change, increase.
(5) increase, increase, increase.

23. The **incorrect** statement with regard to the oxoacids of chlorine HOCl, $HClO_2$, $HClO_3$ and $HClO_4$ is,

(1) The shapes around chlorine in $HClO_2$, $HClO_3$ and $HClO_4$ respectively are angular, pyramidal and tetrahedral.
(2) The oxidation states of chlorine in HOCl, $HClO_2$, $HClO_3$ and $HClO_4$ respectively are +1, +3, +5 and +7.
(3) The acid strength of the oxoacids varies as $HOCl < HClO_2 < HClO_3 < HClO_4$.
(4) All these oxoacids contain at least one double bond.
(5) All these oxoacids contain at least one OH group.

24. The density of an aqueous acidic solution at 25 °C is 1.0 kg dm^{-3}. If the pH of this solution is 1.0, its H^+ concentration in ppm would be,

(1) 0.1 (2) 1 (3) 100 (4) 1000 (5) 10,000

[See page five

25. A 25.0 g sample of polluted air containing ozone (O_3) is treated with an acidic solution containing excess KI. Ozone is converted to O_2 and H_2O during this reaction. The iodine liberated is titrated with 0.002 mol dm^{-3} $Na_2S_2O_3$ solution. Volume of $Na_2S_2O_3$ required was 25.0 cm^3. The mass percent of O_3 in the air sample is, (O = 16)

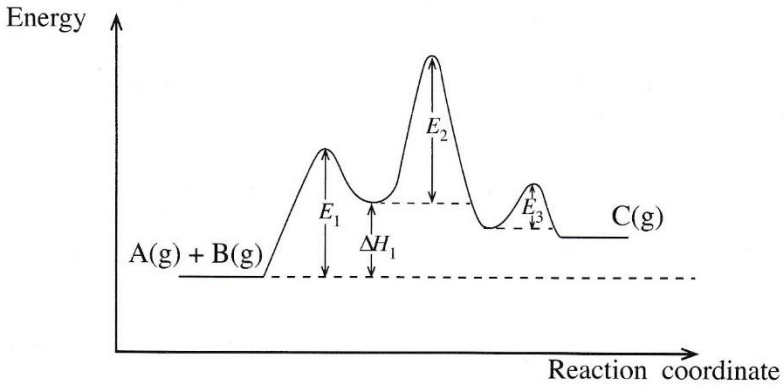
(1) 4.8×10^{-3} (2) 6.4×10^{-3} (3) 9.6×10^{-3} (4) 1.0×10^{-2} (5) 3.2×10^{-2}

26. Which of the following reaction steps is **not** present in the Born-Haber cycle of NaCl(s) formation?

(1) $Na^+(aq) + Cl^-(aq) \longrightarrow NaCl(aq)$ (2) $Na(s) \longrightarrow Na(g)$ (3) $Cl_2(g) \longrightarrow 2Cl(g)$

(4) $Cl(g) + e \longrightarrow Cl^-(g)$ (5) $Na^+(g) + Cl^-(g) \longrightarrow NaCl(s)$

27. Activation energy of the elementary reaction A(g) + B(g) $\longrightarrow$ C(g) is *Ea*. This reaction is catalysed by the metal **M**. The energy diagram of the catalysed reaction is given below.



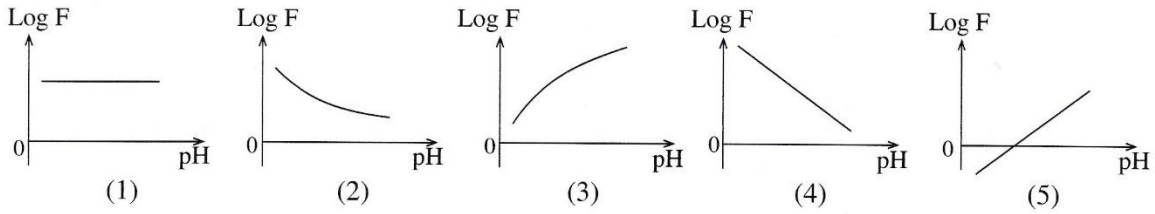
Which of the following is always correct with regard to this reaction?

(1) $Ea < E_1$ (2) $Ea = E_1 + E_2 + E_3 - \Delta H_1$ (3) $Ea < E_1$, $Ea < E_2$ and $Ea < E_3$

(4) $Ea > E_1 + E_2$ (5) $Ea > \Delta H_1 + E_2$

28. For a weak acid, it can be given that $F = \dfrac{\text{Amount of the acid dissociated}}{\text{Amount of the acid undissociated}}$

Which of the following graphs shows the relationship between Log F and pH?



29. Which of the following statements with regard to polymers is correct?

(1) Nylon is an addition polymer.

(2) Teflon is a condensation polymer.

(3) Terelene is a thermosetting addition polymer.

(4) The number of carbon atoms in the repeating unit of natural rubber is 4.

(5) Small covalent molecules are eliminated when monomers combine to form condensation polymers.

30. Two ideal gases that do not react with each other are separated by a valve and kept in a rigid container. This system is maintained at constant temperature and pressure. Which of the following correctly describes the change in Gibbs energy, enthalpy and entropy of the system respectively when the valve is opened?

(1) decreased, decreased, decreased (2) decreased, decreased, increased

(3) decreased, unchanged, increased (4) decreased, increased, increased

(5) increased, increased, increased

● For each of the questions **31** to **40**, one or more responses out of the four responses (*a*), (*b*), (*c*) and (*d*) given is/are correct. Select the correct response/responses. In accordance with the instructions given on your answer sheet, mark

(1) if only (*a*) and (*b*) are correct.
(2) if only (*b*) and (*c*) are correct.
(3) if only (*c*) and (*d*) are correct.
(4) if only (*d*) and (*a*) are correct.
(5) if **any other** number or combination of responses is correct.

Summary of above Instructions

(1)	(2)	(3)	(4)	(5)
Only (*a*) and (*b*) are correct	Only (*b*) and (*c*) are correct	Only (*c*) and (*d*) are correct	Only (*d*) and (*a*) are correct	**Any other** number or combination of responses is correct

31. Which of the following statement/statements is/are correct with regard to simple covalent molecules containing oxygen and sulphur atoms?
(*a*) H_2O shows amphoteric properties.
(*b*) The boiling point of H_2O_2 is higher than the boiling point of H_2O.
(*c*) H_2O_2 can act as an oxidizing agent only in an acidic medium.
(*d*) Both H_2S and SO_2 have the capacity to act only as reducing agents.

32. Which of the following statement/statements is/are correct with regard to hydrocarbons?
(*a*) All hydrocarbons give CO_2 and H_2O when completely reacted with excess O_2.
(*b*) All alkynes react with Grignard reagents to give alkynylmagnesium halides.
(*c*) The boiling point of a branched alkane is higher than the boiling point of an unbranched alkane with the same relative molecular mass.
(*d*) None of the hydrocarbons react with aqueous NaOH.

33. If an endothermic reaction occurs spontaneously at constant temperature and pressure, then,
(*a*) enthalpy of the system decreases. (*b*) entropy of the system increases.
(*c*) enthalpy of the system increases. (*d*) entropy of the system does not change.

34. Which of the following statement/statements is/are correct regarding the precipitation of metal ions by passing $H_2S(g)$ in to their aqueous solutions?
(*a*) When the pressure of $H_2S(g)$ is decreased, the sulphide ion concentration is increased.
(*b*) When the temperature is increased, the sulphide ion concentration is decreased.
(*c*) Addition of $Na_2S(s)$ to the solution, decreases the dissociation of dissolved $H_2S(aq)$.
(*d*) Increase in pH of the solution decreases sulphide ion concentration.

35. Which of the following is/are nucleophilic substitution reaction/reactions?

(*a*) $CH_3\overset{\overset{O}{\|}}{C}-H + HCN \longrightarrow CH_3\overset{\overset{OH}{|}}{C}HCN$

(*b*) $CH_3CH_2OH + PCl_3 \longrightarrow CH_3CH_2Cl$

(*c*) $CH_3\overset{\overset{CH_3}{|}}{C}HCl + NaOH \longrightarrow CH_3\overset{\overset{CH_3}{|}}{C}HOH$

(*d*) $CH_3\overset{\overset{CH_3}{|}}{C}HCH_3 + Cl_2 \xrightarrow{h\upsilon} CH_3\overset{\overset{CH_3}{|}}{\underset{\underset{Cl}{|}}{C}}CH_3$

36. Which of the following statement/statements is/are correct regarding the elevation of carbon dioxid level in the atmosphere?

(*a*) It contributes to the increase in acidity of sea water.
(*b*) It reduces the hardness of water bodies.
(*c*) It strongly absorbs UV radiation coming from the sun.
(*d*) It does not contribute to acid rain.

37. Which of the following statement/statements is/are correct with regard to 3*d*-block elements?

(*a*) Zn has the highest first ionization energy among the 3*d*-block elements.
(*b*) In contrast to the ions of most main group elements (*s* and *p*-block), 3*d*-block metal ions rarel attain the noble gas configuration.
(*c*) Although the electronegativities of 3*d*-block elements are higher than the electronegativitie of the corresponding *s*-block elements, their atomic radii are smaller than the atomic radii o the corresponding *s*-block elements.
(*d*) The 3*d*-block elements that form colourless compounds are Ti and Zn.

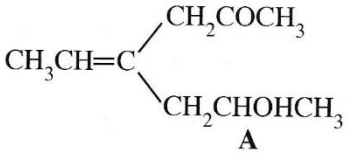
38. Volatile liquids **A** and **B** having saturated vapour pressures P_A° and P_B° $\left(P_A^\circ \neq P_B^\circ\right)$ form an idea solution. A mixture of the liquids **A** and **B** is in equilibrium with their vapour phase, in a close container. When the volume of the container is increased and the equilibrium is re-established a the same temperature, which of the following statement/statements is/are correct?

(*a*) While some amount of **A** and **B** go to the gas phase, the composition of the liquid phase remain unchanged.
(*b*) While some amount of **A** and **B** go to the gas phase, the composition of the gas phase remain unchanged.
(*c*) While some amount of **A** and **B** go to the gas phase, the composition of the liquid phas changes.
(*d*) While some amount of **A** and **B** go to the gas phase, the composition of the gas phas changes.

39. $A(g) \rightarrow B(g)$ is an elementary reaction. Which of the following statement/statements is/are correc regarding the half life of this reaction?

(*a*) Concentration of A is decreased by ⅓ after three half lives.
(*b*) Half life is independent of the concentration of B.
(*c*) Half life decreases when the temperature is increased.
(*d*) Half life of the reaction decreases as the reactant A is consumed.

40. Which of the following statement/statements regarding compound **A** is/are correct?



A

(*a*) **A** exhibits geometric isomerism.
(*b*) **A** does not exhibit optical isomerism.
(*c*) The product obtained when **A** is reacted with pyridinium chlorochromate (PCC) exhibits optical isomerism.
(*d*) The product obtained when **A** is reacted with pyridinium chlorochromate does not exhibit geometric isomerism.

- In question Nos. **41** to **50**, two statements are given in respect of each question. From the Table given below, select the response, out of the responses (1), (2), (3), (4) and (5), that **best** fits the two statements and mark appropriately on your answer sheet.

Response	First Statement	Second Statement
(1)	True	True, and correctly explains the first statement
(2)	True	True, but does **not** explain the first statement correctly
(3)	True	False
(4)	False	True
(5)	False	False

	First Statement	Second statement
41.	Among the halogens, I_2 is a solid whereas Br_2 is a liquid.	London forces become stronger with increase in molecular surface area.
42.	At a given pressure, the spontaneity of the reaction between N_2 and H_2 to give NH_3 drops with increasing temperature.	Entropy change of the reaction between N_2 and H_2 to give NH_3 is negative.
43.	Essential oils are generally extracted from plant materials by steam distillation.	Essential oils have a high solubility in water.
44.	A spontaneous reaction always has a negative Gibbs energy change no matter what the conditions are.	Gibbs energy change can be used to predict the direction of a reaction only under constant temperature and constant pressure conditions.
45.	Solubility of 1-butanol in water is less than the solubility of methanol in water.	The solubility of alcohols in water decreases as the size of the non-polar alkyl group increases relative to the polar OH group.
46.	The reaction, $CH_3-CH=CH_2 \xrightarrow{HBr} CH_3-\underset{Br}{CH}-CH_3$ is a nucleophilic addition reaction.	A secondary carbocation is formed as a reaction intermediate in the following reaction. $CH_3-CH=CH_2 \xrightarrow{HBr} CH_3-\underset{Br}{CH}-CH_3$
47.	Coke is used in several industrial processes.	Coke is only used industrially as a fuel.
48.	The carbonyl carbon atom of a ketone and the other atoms bonded to it lie in the same plane.	The carbonyl carbon atom of a ketone is sp^2 hybridized.
49.	Any two ideal gases have the same average kinetic energies at the same temperature.	At a given temperature, the average speed of gas molecules adjust according to their masses.
50.	Although CFC contribute to ozone layer depletion, the contribution from HFC is negligible.	HFC undergoes complete decomposition before reaching the upper atmosphere.

* * *

[See page nine

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்

අ.පො.ස. (උ.පෙළ) විභාගය/ க.பொ.த. (உயர் தர)ப் பரீட்சை - 2019

පැරණි නිර්දේශය/ பழைய பாடத்திட்டம்

විෂයය අංකය
பாட இலக்கம் 02

විෂයය
பாடம் **Chemistry**

ලකුණු දීමේ පටිපාටිය/புள்ளி வழங்கும் திட்டம்

I පත්‍රය/பத்திரம் I

ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.
01.	2	11.	4	21.	2	31.	1 or 5	41.	1
02.	5	12.	2	22.	2	32.	4	42.	1
03.	3	13.	2	23.	4	33.	2	43.	3
04.	all	14.	2 or 5	24.	3	34.	2	44.	4
05.	5	15.	2	25.	1	35.	2	45.	1
06.	1	16.	5	26.	1	36.	4	46.	4
07.	1	17.	4	27.	5	37.	5	47.	3
08.	2	18.	4	28.	5	38.	3	48.	1
09.	2	19.	3	29.	5	39.	2	49.	1
10.	4	20.	3	30.	3	40.	4	50.	all

✪ විශේෂ උපදෙස්/ விசேட அறிவுறுத்தல் :

එක් පිළිතුරකට/ ஒரு சரியான விடைக்கு 01 ලකුණු බැගින්/புள்ளி வீதம்

මුළු ලකුණු/மொத்தப் புள்ளிகள் 1 × 50 = 50

PART A — STRUCTURED ESSAY

*Answer **all four** questions on this paper itself. (Each question carries **10** marks.)*

(*a*) The following questions are related to the elements of the second row in the Periodic Table. Write the **symbol** of the element in the space provided in answering parts (i) to (vi).

(i) Identify the element that has the highest electronegativity (disregard the noble gas). ..... F

(ii) Identify the element that has an allotrope which conducts electricity. C

(iii) Identify the element that forms the monoatomic ion largest in size (this should be a stable ion). .....N.....

(iv) Identify the element that has **no** *p* electrons but has a stable *s* configuration. ..... Be

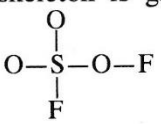
(v) Identify the element that has the highest first ionization energy. .....Ne.....

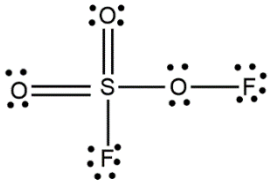
(vi) Identify the element that forms mostly electron deficient trigonal planar covalent compounds. .....B.....

Note: If name is given instead of symbols do not award marks (04 X 6 = 24)

1(a): 24 marks

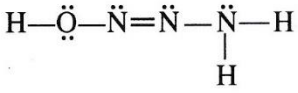
(*b*) (i) Draw the **most** acceptable Lewis structure for the molecule SO_3F_2. Its skeleton is given below.

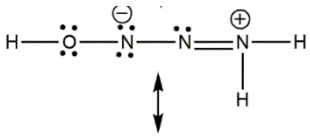




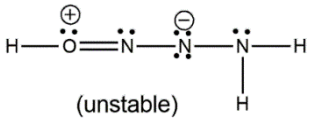
(06)

(ii) The most stable Lewis structure for the molecule H_3N_3O is shown below. Draw **two** more Lewis structures (resonance structures) for this molecule. Write 'unstable' under the more unstable structure drawn by you.





(04)



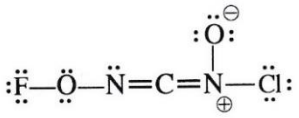
(unstable)

(04)

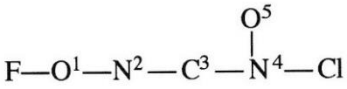
(02)

(iii) Based on the Lewis structure given below, state the following regarding the C, N and O atoms given in the table.

I. VSEPR pairs around the atom
II. electron pair geometry around the atom
III. shape around the atom
IV. hybridization of the atom



The atoms are numbered as follows.



		O^1	N^2	C^3	N^4
I	VSEPR pairs	4	3	2	3
II	electron pair geometry	tetrahedral	trigonal planar	linear	trigonal planar
III	shape	angular / V / bent	angular / V / bent	linear	trigonal planar
IV	hybridization	sp^3	sp^2	sp	sp^2

(01 X 16 = 16)

(iv) Identify the atomic/hybrid orbitals involved in the formation of the following σ bonds in the Lewis structure given in part (iii) above. (Numbering of atoms is as in part (iii).)

I.	$F—O^1$	F	$2p$ **OR** sp^3	O^1	sp^3
II.	$O^1—N^2$	O^1	sp^3	N^2	sp^2
III.	$N^2—C^3$	N^2	sp^2	C^3	sp
IV.	$C^3—N^4$	C^3	sp	N^4	sp^2
V.	$N^4—O^5$	N^4	sp^2	O^5	$2p$ **OR** sp^3
VI.	$N^4—Cl$	N^4	sp^2	Cl	$3p$ **OR** sp^3

(01 X 12 = 12)

(v) Identify the atomic orbitals involved in the formation of the following π bonds in the Lewis structure given in part (iii) above. (Numbering of atoms is as in part (iii).)

I.	$N^2—C^3$	N^2	$2p$	C^3	$2p$
II.	$C^3—N^4$	C^3	$2p$	N^4	$2p$

(01 X 4 = 04)

(vi) I. How are the two double bonds oriented in the Lewis structure given in part (iii)?

Double bonds are perpendicular to each other **(02)**

OR

Sigma bonds linear, π bonds perpendicular **(01 + 01)**

II. Give an example of a molecule/ion that has a similar orientation of double bonds.

CO_2, NO_2^+ , CN_2^{2-} , N_3^- **(02)**

Note: Your example should not contain more than 3 atoms.
The elements in your example should be restricted to the first and second periods of the Periodic Table.

1(b): 52 marks

(c) (i) An atomic orbital is described by three quantum numbers n, l and m_l.

Write the appropriate quantum numbers and the name of the atomic orbital in the boxes given below.

	n	l	m_l	atomic orbital
I.	3	1	+1	$3p$
II.	3	2	–2	3d
III	2	0	0	$2s$

(01 X 6 = 06)

(ii) Arrange the following in the **increasing** order of the property indicated in parenthesis. (Reasons are **not** required.)

I. LiF, LiI, KF (melting point)

LiI < LiF < KF

II. NO_2^-, NO_4^{3-}, NF_5 (stability)

NF_5 < NO_4^{3-} < NO_2^-

III. NOCl, $NOCl_3$, NO_2F (N—O bond distance)

NOCl < NO_2F < $NOCl_3$

(06 X 3 = 18)

1(c): 24 marks

2. (*a*) **X** is an *s*-block element in the Periodic Table. The first, second and third ionization energies of **X**, in kJ mol^{-1} are 738, 1451 and 7733 respectively. **X** reacts slowly with hot water, liberating $H_2(g)$ and forming its hydroxide. The hydroxide is basic. **X** also liberates $H_2(g)$ on reaction with dilute acids. **X** burns in air with a bright white light. A compound of **X** is used to remove acidic gases from polluted air.

(i) Identify **X**. **X**: Mg ...**OR**.... Magnesium **(07)**

(ii) Write the ground state electronic configuration of **X**. $1s^2 2s^2 2p^6 3s^2$ **(04)**

(iii) Write the chemical formulae of the **two** compounds formed when **X** burns in air.

MgO and Mg_3N_2 **(03 + 03)**

Note: If X has been identified correctly marks can be awarded for XO and X_3N_2

(iv) Consider the given compounds of the elements in the group in the Periodic Table to which **X** belongs. In the given boxes, write whether the indicated property **increases** or **decreases** down the group.

I. Solubility of sulphates in water — Decreases **(03)**

II. Solubility of hydroxides in water — Increases **(03)**

III. Thermal stability of metal carbonates — Increases **(03)**

Give reasons for your answer in III.

Cation size increases down the group. Charge same. **(03)**
OR
Charge density decreases down the group. **(03)**
Therefore, polarizing power decreases down the group. **(02)**
Hence, decomposition of the carbonates become difficult on going down the group. **(03)**

(v) Identify the element in the *s*-block of the Periodic Table, which reacts in a similar manner to **X** with $H_2(g)$, $O_2(g)$ and $N_2(g)$, but does **not** belong to the same group as **X**.

Lithium **OR** Li **(04)**

(vi) Write the chemical formula of the compound of **X** used to remove acidic gases from polluted air.

MgO **(04)**

(vii) Write the balanced chemical equation for the reaction between the compound of **X** identified in (vi) above and SO_2.

$MgO + SO_2 \rightarrow MgSO_3$ **(04)**

(viii) **X** is a component of a well-known reagent used in organic chemistry. Give the **name** of this reagent.

Grignard reagent **(04)**

Note: If X is incorrect, do not award marks for (a)(ii) – (iv).

2(a): 50 marks

(*b*) Test tubes labelled **A** to **E** contain aqueous solutions of $Na_2S_2O_3$, Na_2CO_3, KNO_2, KBr and Na_2S (not in order). The characteristics of the solutions obtained and gases evolved on addition of dil. HCl (warming if required) to each of the test-tubes **A** to **E** are given in the table below.

Test-tube	Appearance of solution	Gas
A	colourless	colourless and odourless
B	colourless	reddish-brown with a pungent odour
C	colourless	colourless with a rotten egg odour
D	turbid	colourless with a pungent odour
E	colourless	not evolved

(i) Identify the solutions in each of the test-tubes **A** to **E**.

A : ...Na_2CO_3..... **C** : .....Na_2S..... **E** : .....KBr.....

B : ....KNO_2..... **D** : .....$Na_2S_2O_3$..... **(04 X 5 = 20)**

(ii) Write balanced chemical equations for the reactions that take place in test-tubes **A**, **B**, **C** and **D**.

In A: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

In B: $KNO_2 + HCl \rightarrow KCl + HNO_2$ **OR**

$2KNO_2 + 2HCl \rightarrow 2KCl + NO_2 + NO + H_2O$ **OR**

$3KNO_2 + 3HCl \rightarrow 3KCl + HNO_3 + 2NO + H_2O$ **OR**

$3KNO_2 + 3HCl + O_2 \rightarrow 3KCl + HNO_3 + H_2O + 2NO_2$ **OR**

$4KNO_2 + 4HCl + O_2 \rightarrow 4KCl + 4NO_2 + 2H_2O$

In C: $Na_2S + 2HCl \rightarrow 2NaCl + H_2S$

In D: $Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$ **(04 X 4 = 16)**

(iii) Write a chemical test to identify each of the gases evolved in **A**, **C** and **D**.

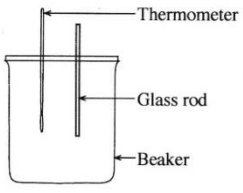
Note: Observations are also required.

In A: **(CO_2)** Pass through a solution of $Ca(OH)_2$. **(02)**
Solution turns milky. On continued passing, milky solution becomes colourless. **(02)**
In C: **(H_2S)** Test with filter paper moistened with lead acetate. **(03)**
Filter paper turns black. **(02)**
OR
Test with filter paper moistened with cadmium acetate. **(03)**
Filter paper turns yellow. **(02)**
OR
Pass through a solution of acidified $KMnO_4$. **(03)**
(Purple) solution becomes colourless, but solution is not clear (cloudy) **(02)**
OR
Pass through a solution of acidified $K_2Cr_2O_7$. **(03)**
(Orange) solution becomes green, but solution is not clear (cloudy) **(02)**
In D: **(SO_2)** Pass through a solution of acidified $KMnO_4$. **(03)**
(Purple) solution becomes colourless. **(02)**
OR
Pass through a solution of acidified $K_2Cr_2O_7$./Test with filter paper moistened with acidified $K_2Cr_2O_7$ **(03)**
(Orange) solution becomes green. / Filter paper turns green. **(02)**
OR
Pass through a solution of $Ca(OH)_2$. **(03)**
Solution turns milky. On continued passing, milky solution becomes colourless. **(02)**
OR
Test with moist coloured flower petals. **(03)**
Petals are decolourized /bleached. **(02)**

Note: Identification in (b)(i) must be correct for award of marks for respective answers in (b)(ii) & (b)(iii).

2(b): 50 marks

3. The set up shown in the figure was used to calculate the heat change associated with the dissolution of MX(s) in water. 100.00 cm^3 of distilled water was added to the beaker. The initial temperature of distilled water was measured to be 25.0 °C. Then 0.10 mol of MX(s) was added to the water and stirred continuously. It was observed that the temperature of the solution decreased gradually. The lowest temperature measured was 17.0 °C. The amount of water used was sufficient to completely dissolve MX(s). Density and specific heat capacity of water are 1.00 g cm^{-3} and 4.20 J g^{-1} $°C^{-1}$ respectively. Assume that the density and the specific heat capacity of water are not changed due to the dissolution of MX(s).



(i) Calculate the amount of heat that should be supplied to bring the system (solution) back to 25.0 °C.

q = m s ΔT (**OR** q = m c ΔT) **(5)**

= 100.00 cm^3 x 1.0 g cm^{-3} x 4.2 J $^0C^{-1}$ g^{-1} x (25.0 - 17.0) ^0C **(4+1)+(4+1)+(4+1)+(4+1)**

= 3360 J **(4+1)**

(ii) Is the dissolution of MX(s) in water an endothermic or exothermic process? Explain your answer.

Dissolution of MX(s) has absorbed heat (**OR** temperature of water decreased) **(2)**

Process is endothermic **(2)**

(iii) Calculate the enthalpy change (in kJ mol^{-1}) associated with reaction

$MX(s) + H_2O(l) \rightarrow M^+(aq) + X^-(aq)$.

$\Delta H = \frac{3360 J}{0.10 mol}$ **(4+1)+(4+1)**

= 33.6 kJ mol^{-1} (**OR** 33600 J mol^{-1}) **(4+1)**

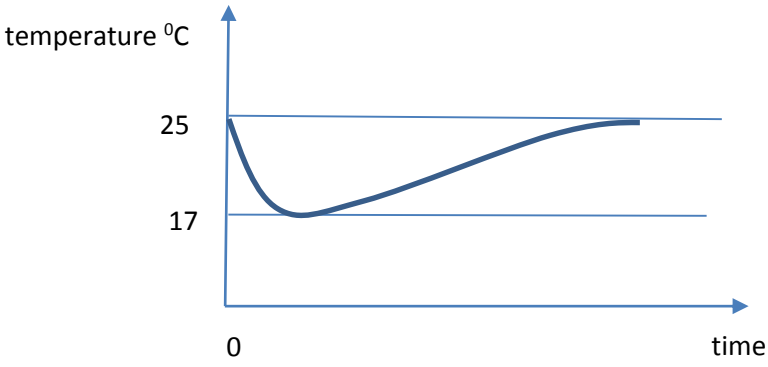
(iv) If this experiment was conducted using 200.00 cm^3 of water, would you expect the temperature change to be larger than the above value? Explain your answer.

No (**OR** the temperature change would be smaller) **(2)**

Amount of heat (q) is constant, the mass (m) has increased. Therefore, the temperature change (ΔT) would be smaller (**OR** More water is available to release heat) **(2)**

(v) Show the variation of temperature of the system (solution) by drawing the temperature – time curve.

Note: Eventually the system reaches the room temperature (25.0 °C).



Curve starts from t=0 (or the time of salt addition is marked)	**(2)**
Curve starts from 25 ^{0}C	**(2)**
Curve goes down to 17 ^{0}C	**(2)**
Curve has the correct shape	**(4)**

(vi) State one experimental error, that could occur during this experiment.

Conduction of heat to the solution from the beaker **(2)**

and surrounding **(2)**

(vii) Gibbs energy change (ΔG) for the dissolution of MX(s) in water at the temperature of 25.0 °C and pressure of 1.0 atm was calculated to be –26.0 kJ mol^{-1}. Calculate the entropy change (ΔS) of dissolution of MX(s) in water at 25.0 °C using the enthalpy change calculated above.

$\Delta G = \Delta H - T\Delta S$ **(No marks for $\Delta G^0 = \Delta H^0 - T\Delta S^0$)** **(5)**

$\Delta S = \frac{\Delta H - \Delta G}{T}$

$= \frac{33.6 kJ mol^{-1} - (-26.0 kJ mol^{-1})}{298 K}$ **(4+1)+(4+1)+(4+1)**

= 200 J mol^{-1} K^{-1} **(4+1)**

(viii) Would you expect the solubility of MX(s) to increase or decrease with increasing temperature? Give reasons for your answer.

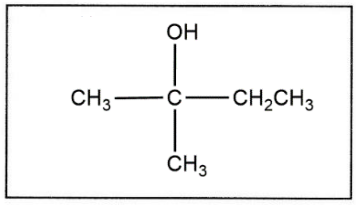
Solubility of MX(s) in water increases with increasing temperature **(4)**

because ΔG becomes more negative. **(4)**

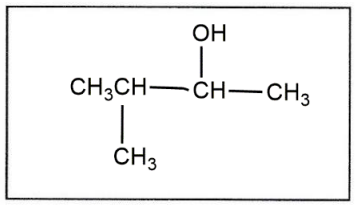
(**OR** because the dissolution of MX(s) is endothermic)

4. (*a*) Compounds **A** and **B** both have the same molecular formula $C_5H_{10}O$. Both **A** and **B** give orange/red precipitates with 2,4-dinitrophenylhydrazine. When **A** and **B** are reacted separately with $NaBH_4$ in methanol, compound **A** gives **C** and compound **B** gives **D**. When **C** is heated with Al_2O_3, two alkenes **E** (C_5H_{10}) and **F** (C_5H_{10}) are formed. When **E** and **F** are reacted separately with conc. H_2SO_4 and the products obtained are hydrolysed, compound **E** gives **G**, while compound **F** gives **H**. **G** gives a turbidity immediately with the Lucas reagent. **H** also gives a turbidity with the Lucas reagent but not immediately.

(i) Draw the structures of **G** and **H**.



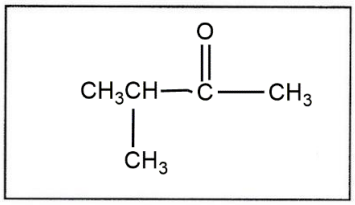
G



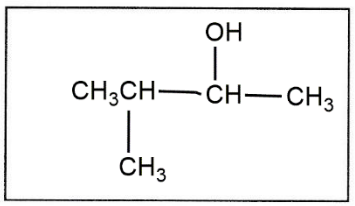
H

(05 x 2 = 10)

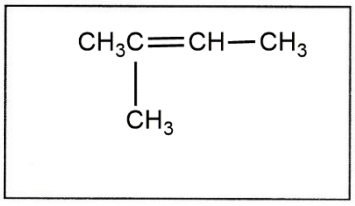
(ii) Draw the structures of **A**, **C**, **E** and **F**.



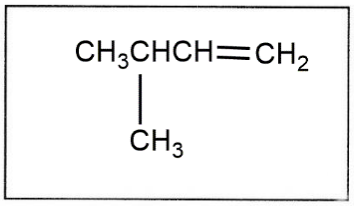
A



C



E

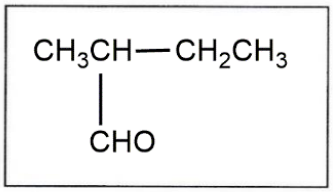


F

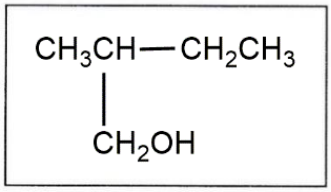
(05 x 4 = 20)

When heated with Al_2O_3, **D** gives alkene **I** (C_5H_{10}). When **I** is reacted with conc. H_2SO_4 and the product obtained hydrolysed, **G** is obtained.

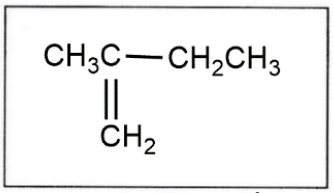
(iii) Draw the structures of **B**, **D** and **I**.



B



D



I

(05 x 3 = 15)

Note : 1. Mark A-I Independently

2. If the correct structure is given for either C or H . award full marks for both C and H (05x2 = 10)

(iv) Describe a test/reaction to distinguish between **A** and **B**.

B gives

Tollens reagent	- Silver mirror
Fehlings solution	- Red colouration
Acidic $K_2Cr_2O_7$	- Green colouration
Dilute $KMnO_4$ solution	- Decolourization

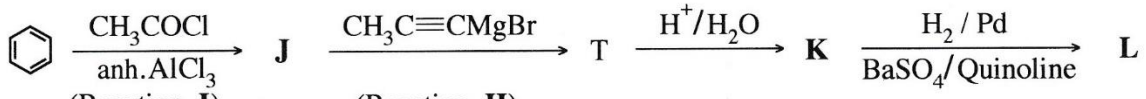
(05)

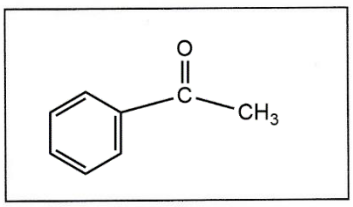
(Any one)

Note: Marks awarded only if A and B are correct.

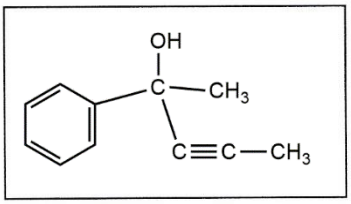
4(a): 50 marks

(*b*) (i) Give the structures of **J**, **K**, **L** and **M** in the following reaction sequences.

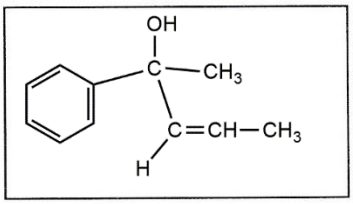




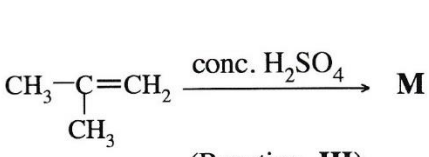
J

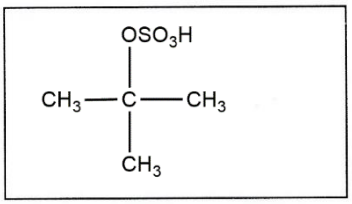


K



L





M

(05 x 4 = 20)

(ii) Selecting from the list given below, write the type of reaction taking place in reactions **I**, **II** and **III**.

Nucleophilic Addition, Nucleophilic Substitution, Electrophilic Addition, Electrophilic Substitution, Elimination

Reaction I - Electrophilic substitution

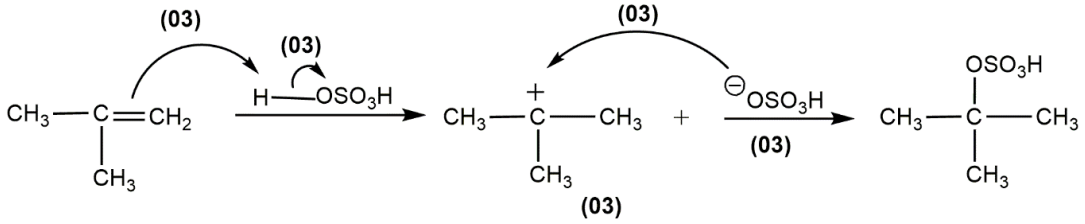
Reaction II - Nucleophilic addition

Reaction III - Electrophilic addition

(05 x 3 = 15)

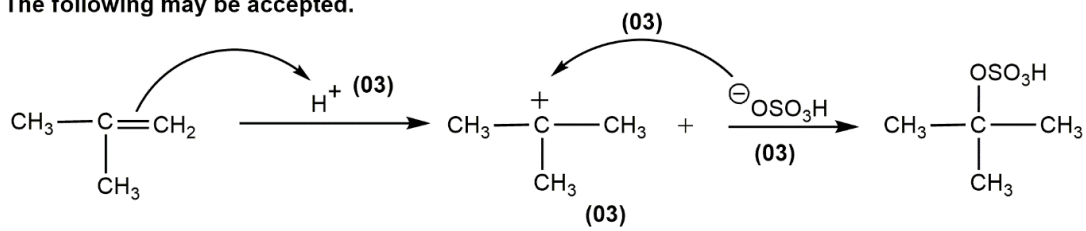
Note : Marks awarded only if each of the reactions I, II and III correct as given in the marking scheme

(iii) Using your knowledge of the mechanism of the reaction between alkenes and HBr, give the mechanism of reaction **III**.



(15 marks)

The following may be accepted.



(12 marks)

4(b): 50 marks

PART B — ESSAY

*Answer **two** questions only. (Each question carries **15 marks**.)*

5. (*a*) A titration between the mono acidic weak base **B** (0.15 mol dm^{-3}) and HCl (0.10 mol dm^{-3}) was carried out using a suitable indicator as described below.

The HCl solution (25.00 cm^3) was kept in the titration flask and the weak base **B** was added using a burette. The dissociation constant, K_b of the weak base at 25 °C is 1.00×10^{-5} mol dm^{-3}. All the experiments were conducted at 25 °C.

(i) Calculate the pH of the acid solution in the titration flask, before the addition of the base, **B**.

pH of the HCl solution.

pH = - log[H^+] (**2**)

= - log(0.1)

= 1.0 (**2+1**)

(ii) Calculate the pH of the solution in the titration flask, after the addition of 10.00 cm^3 of the solution of **B**. Can the solution in the titration flask act as a buffer solution? Explain your answer.

pH after addition of 10.00 cm^3 of B solution.

$[H^+] = \frac{0.1 \text{ mol } dm^{-3} \times 25.00 \text{ } cm^3 - 0.15 \text{ mol } dm^{-3} \times 10.00 \text{ } cm^3}{35.00 \text{ } cm^3}$ (**4+1**)

= 0.028 mol dm^{-3}

pH = 1.5 (**OR** 1.6) (**4+1**)

No **OR** this solution cannot act as a buffer solution. (**3**)

Only protonated base (conjugate acid) is present (No unreacted or unprotonated base) (**3**)

Note : If correctly explained using the addition of H^+ and OH^-, award full marks

(iii) Calculate the volume of the weak base solution required to reach the equivalence point.

Volume of base required to reach equivalence point.

$V = \frac{0.1 \text{ mol } dm^{-3} \times 25.00 \text{ } cm^3}{0.15 \text{ mol } dm^{-3}}$ (**4+1**)

= 16.66 cm^3

(16.67 cm^3 **OR** the answer reported to one decimal place is also accepted) (**4+1**)

(iv) Another 10.00 cm^3 volume of the weak base was added to the titration flask after reaching the equivalence point. Calculate the pH of the solution in the titration flask.

pH after addition of 10.00 cm^3 of base after reaching the equivalence point.

Weak base dissociates according to,

B (aq) + H_2O(l) ⇌ BH^+(aq) + OH^-(aq) (**2**)

$K_b = \frac{[BH^+(aq)][OH^-(aq)]}{[B(aq)]}$ (**4**)

OR

$$pOH = pK_b + log\left(\frac{[BH^+(aq)]}{[B(aq)]}\right)$$

Note : Physical states are required for the award of marks

Assuming that the amount dissociated is negligible, (**2**)

Concentration of the weak base [B(aq)] = $\frac{0.15 \text{ mol dm}^{-3} \times 10.00 \text{ cm}^3}{(25.00 + 16.66 + 10.00) \text{ cm}^3}$ (**4+1**)

Concentration of the protonated weak base [BH^+(aq)] = $\frac{0.15 \text{ mol dm}^{-3} \times 16.66 \text{ cm}^{-3}}{(25.00 + 16.66 + 10.00) \text{ cm}^3}$ (**4+1**)

$$pOH = -\log(1 \times 10^{-5}) + \log\left(\frac{0.15\ mol\ dm^{-3} \times 16.66\ cm^3}{0.15\ mol\ dm^{-3} \times 10.00\ cm^3}\right)$$ (**4+1**)

pOH = 5.0 + 0.221 = 5.221

pH = 8.78 (**OR** 8.7 **OR** 8.9 **OR** 9) (**4+1**)

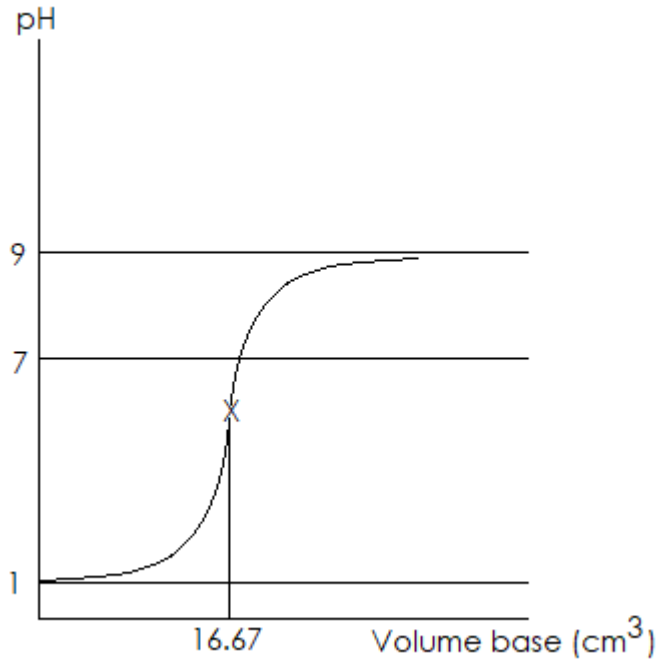
(v) Can the solution obtained in (iv) above act as a buffer solution? Explain your answer.

Yes **OR** it can act as a buffer solution. (**3**)

The solution in the titration flask contains the protonated base (conjugate acid) and unreacted base. (**3**)

Note : If correctly explained using the addition of H^+ and OH^-, award full marks

(vi) Sketch the variation in pH of the mixture in the titration flask with the volume of the weak base solution added (titration curve). Label the axes, indicate pH on the *y*-axis and the volume of weak base solution added on the *x*-axis. Mark the equivalence point approximately. [Calculation of pH at equivalence point is not expected.]



Curve starts at pH=1 and reaches pH=9 and has the correct shape (**4**)

Equivalence volume marked (**2**)

Equivalence pH (between pH = 5 and pH=7) marked (**2**)

Axes are labeled (with units where applicable) (**1+1**)

5 (a): 75 marks

(*b*) The following two experiments were carried out at a constant temperature using the volatile liquids **C** and **D** which form an ideal solution.

Experiment I: The liquids **C** and **D** were introduced in to an evacuated rigid container and allowed to reach equilibrium. When the system was at equilibrium, it was observed that the mole fractions of **C** and **D** in the liquid phase (L_I) were 0.3 and 0.7 respectively. Total pressure in the container was 2.70×10^4 Pa.

Experiment II: This experiment was conducted using different amounts of **C** and **D**. When the equilibrium was established, it was observed that the mole fractions of **C** and **D** in the liquid phase (L_{II}) were 0.6 and 0.4 respectively. Total pressure of the container was 2.40×10^4 Pa.

(i) Give the relationship between the partial pressure of **C** in the vapour phase (P_C), its saturated vapour pressure $\left(P_C^\circ\right)$ and its mole fraction in the liquid phase (X_C) in the form of an equation. This equation states a commonly used law in physical chemistry. Write the name of the law.

$P_C = x_C P^0_C$ **(Award marks only for these symbols)** **(5)**

Raoult's law **(4)**

(ii) Calculate the saturated vapour pressures of **C** and **D**.

Experiment I

2.7×10^4 Pa $= 0.3\ P^0_C + 0.7\ P^0_D$ ---(1) **(4+1)**

Experiment II

2.4×10^4 Pa $= 0.6\ P^0_C + 0.4\ P^0_D$ ---(2) **(4+1)**

(1)x2-(2)

$P^0_D = 3.0 \times 10^4$Pa **(4+1)**

$P^0_C = (2.4 \times 10^4\ \text{Pa} - 0.4 \times 3.0 \times 10^4\text{Pa})/0.6$

$= 2.0 \times 10^4$ Pa **(4+1)**

(iii) Calculate the mole fractions of **C** and **D** in the vapour phase (V_I) of experiment I.

Mole fractions in the gas phase (experiment I, V_I)

$X^g_{C,I} = \dfrac{0.3 \times 2.0 \times 10^4\ \text{Pa}}{2.7 \times 10^4\text{Pa}}$ **(1+1)**

$= 0.2$ (**OR** 0.22 **OR** 2/9) **(1+1)**

$X^g_{D,I} = 1 - 0.2$ (**OR** 0.22) **(1+1)**

$= 0.8$ (**OR** 0.78 **OR** 7/9) **(1+1)**

(iv) Calculate the mole fractions of **C** and **D** in the vapour phase (V_{II}) of experiment II.

Mole fractions in the gas phase (experiment II, V_{II})

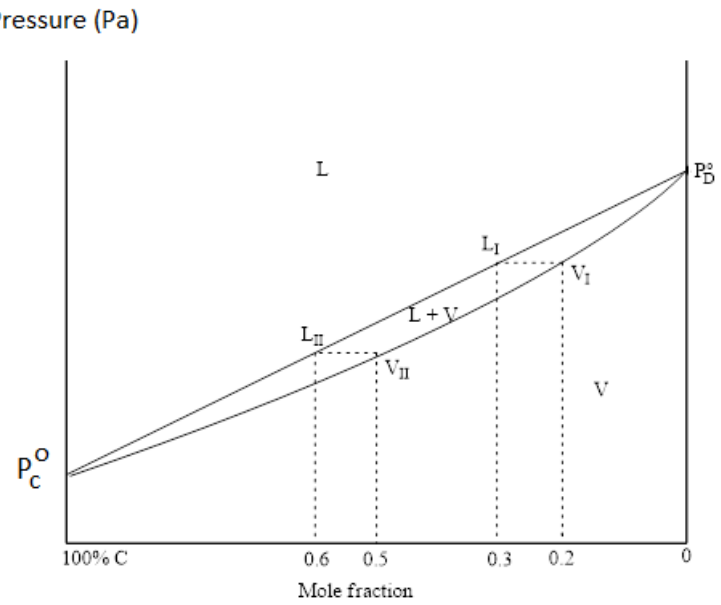
$X^g_{C,II} = \dfrac{0.6 \times 2.0 \times 10^4\ \text{Pa}}{2.4 \times 10^4\text{Pa}}$ **(1+1)**

$= 0.5$ **(1+1)**

$X^g_{D,II} = 1 - 0.5$ **(1+1)**

$= 0.5$ **(1+1)**

(v) Show the compositions of liquid and vapour phases (L_I, L_{II}, V_I and V_{II}) and relevant pressures in the above two experiments on a pressure-composition phase diagram drawn at constant temperature.



L=liquid, V= vapour

Note :Graph could also be drawn by reversing the increasing direction of the mole fraction of C. Mark accordingly

Axes labeled (with appropriate units where applicable)	(**2+2**)
P^0_C and P^0_D marked	(**2+2**)
Line & curve (starts and ends at correct pressures)	(**2+2**)
Phases at equilibrium in each region identified	(**2+2+2**)
point L_I marked at X_C = 0.3	(**2**)
point L_{II} marked at X_c = 0.6	(**2**)
point V_I marked at X_C = 0.2	(**2**)
point V_{II} marked at X_C = 0.5	(**2**)
points L_I and V_I are at the same level	(**2**)
points L_{II} and V_{II} are at the same level	(**2**)

Note : No marks for temperature composition phase diagram

5 (b): 75 marks

6. (*a*) An organic solvent (org-1) and water(aq) are immiscible and form a biphasic system. Partition coefficient for the distribution of **X** between org-1 and water at temperature T is, $K_D = \frac{[\mathbf{X}]_{org\text{-}1}}{[\mathbf{X}]_{aq}} = 4.0$

An amount of 0.50 mol of **X** was added to a system containing 100.00 cm^3 of org-1 and 100.00 cm^3 of water. The system was allowed to reach equilibrium at temperature T.

(i) Calculate the concentration of **X** in org-1.

Calculation of $[X]_{org\text{-}1}$

$$K_D = \frac{[X]_{org-1}}{[X]_{aq}} = 4.0$$

V= volume, x= amount (mols) in aqueous phase.

$$K_D = \frac{\frac{0.5\ mol - x}{V}}{\frac{x}{V}} = 4.0$$ **(No marks for substitution in mols)** **(4+1)**

x = 0.1 mol **(4+1)**

$$[x]_{org-1} = \frac{0.4\ mol}{100\times10^{-3} dm^3} = 4.0 \text{ mol dm}^{-3}$$ **(4+1)**

(ii) Calculate the concentration of **X** in water.

$$[x]_{aq} = \frac{0.1\ mol}{100\times10^{-3} dm^3} = 1.0 \text{ mol dm}^{-3}$$ **(4+1)**

6 (a): 20 marks

(*b*) The compound **Y** is soluble only in the aqueous phase. In the aqueous phase, **X** and **Y** react to form **Z**. The presence of **Y** and **Z** does not affect the distribution of **X** between org-1 and water.

A series of biphasic systems containing org-1 and water were prepared. Then different amounts of **X** were distributed in the biphasic systems and the systems were allowed to reach equilibrium. The initial rate of the reaction between **X** and **Y** in the aqueous phase was measured after adding **Y** into the aqueous phase of these biphasic systems. Results of these experiments conducted at temperature T are given in the table.

Experiment Number	Volume of water (cm^3)	Volume of org-1 (cm^3)	Total amount of X added (mol)	Total amount of Y added (mol)	Initial rate of the reaction (mol dm^{-3} s^{-1})
1	100.00	100.00	0.05	0.02	2.00×10^{-6}
2	100.00	100.00	0.10	0.04	1.60×10^{-5}
3	50.00	50.00	0.25	0.02	4.00×10^{-4}

Orders of the reaction with respect to **X** and **Y** are m and n respectively. The rate constant of the reaction at temperature T is k.

(i) Given that the concentrations of **X** and **Y** in the aqueous phase are $[\mathbf{X}]_{aq}$ and $[\mathbf{Y}]_{aq}$ respectively, write the rate expression for the reaction in terms of $[\mathbf{X}]_{aq}$, $[\mathbf{Y}]_{aq}$, m, n and k.

$Rate = k\,[X]^m_{aq}\,[Y]^n_{aq}$ **OR** $\frac{-\Delta[X]_{aq}}{\Delta t} = k[X]^m_{aq}\,[Y]^n_{aq}$ **OR** $\frac{-\Delta[Y]_{aq}}{\Delta t} = k[X]^m_{aq}\,[Y]^n_{aq}$ **(10)**

(ii) Calculate the initial concentration of **X** in the aqueous phase in each experiment.

Let x be the amount (moles) of X in the aqueous phase and n_x be the total amount (moles) of X added.

Equal volumes of org-1 and water are used in all the experiments.

$$[X]_{aq} = \frac{n_X}{5 \times V_{aq}}$$

Experiment	$[X]_{aq}$/mol dm^{-3}	
1	0.1	**(4)**
2	0.2	**(4)**
3	1.0	**(4)**

(iii) Calculate the initial concentration of **Y** in the aqueous phase in each experiment.

Let n_Y be the total amount of Y (moles) added.

Y is only soluble in the aqueous phase.

$$[Y]_{aq} = \frac{n_Y}{V_{aq}}$$

Experiment	$[Y]_{aq}$/mol dm^{-3}	
1	0.2	**(4)**
2	0.4	**(4)**
3	0.4	**(4)**

(iv) Calculate the orders *m* and *n* of the reaction with respect to **X** and **Y** respectively.

2.00×10^{-6} mol dm^{-3} s^{-1} = k (0.1 mol dm^{-3})^m (0.2 mol dm^{-3})^n ---(1) (**10 +2**)

1.60×10^{-5} mol dm^{-3} s^{-1} = k (0.2 mol dm^{-3})^m (0.4 mol dm^{-3})^n ---(2) (**10 +2**)

4.00×10^{-4} mol dm^{-3} s^{-1} = k (1.0 mol dm^{-3})^m (0.4 mol dm^{-3})^n ---(3) (**10 +2**)

Finding of the order m

From (2)/(3)

$$\frac{1.60 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}}{4.00 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}} = \frac{k\,(0.2 \text{ mol dm}^{-3})^m\,(0.4 \text{ mol dm}^{-3})^n}{k\,(1.0 \text{ mol dm}^{-3})^m\,(0.4 \text{ mol dm}^{-3})^n}$$ **(5)**

$0.04 = (0.2)^m$

m = 2 (**4+1**)

Finding of the order n

From (3)/(1)

$$\frac{4.00 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}}{2.00 \times 10^{-6} \text{ mol dm}^{-3} \text{ s}^{-1}} = \frac{k\,(1.0 \text{ mol dm}^{-3})^m\,(0.4 \text{ mol dm}^{-3})^n}{k\,(0.1 \text{ mol dm}^{-3})^m\,(0.2 \text{ mol dm}^{-3})^n}$$ (**5**)

$200 = 10^2\,(2)^n$

n = 1 (**4+1**)

(v) Calculate the rate constant of the reaction.

Rate constant

From (1)

$k = \frac{2.00 \times 10^{-6} \text{ mol dm}^{-3} \text{ s}^{-1}}{(0.1 \text{ mol dm}^{-3})^2 (0.2 \text{ mol dm}^{-3})^1}$ **(4+1)**

$= 1.0 \times 10^{-3} \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ **(4+1)**

(vi) An experiment is designed to study the effect of temperature on the reaction rate using the partition coefficient given above.

Is this a suitable experiment to study the effect of temperature on the rate of the reaction? Explain your answer.

Not suitable **(2)**

Partition coefficient depends on temperature. **(3)**

6 (b): 105 marks

(*c*) The organic solvent org-2 and water are also immiscible and form a biphasic system. **X** (0.20 mol) was added to a system containing 100.00 cm^3 of org-2 and 100.00 cm^3 of water and allowed to reach equilibrium at the temperature T. Then **Y** (0.01 mol) was added to the aqueous phase and the initial rate of the reaction was measured. **Y** does not dissolve in org-2. The initial rate of the reaction between **X** and **Y** in the aqueous phase was found to be 6.40×10^{-7} mol dm^{-3} s^{-1}.

Calculate the partition coefficient $\frac{[\mathbf{X}]_{\text{org-2}}}{[\mathbf{X}]_{\text{aq}}}$ for the distribution of **X** between org-2 and water.

$[\mathbf{X}]_{\text{org-2}}$ is the concentration of **X** in the org-2 phase.

Reaction takes place in the aqueous medium. Therefore, the rate constant is the same. **(5)**

$Rate = k\,[X]^2_{aq}\,[Y]_{aq}$

$6.40 \times 10^{-7} \text{ mol dm}^{-3} \text{ s}^{-1} = 1.00 \times 10^{-3} \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1} [X]^2_{aq}$ 0.1 mol dm^{-3} **(4+1)**

$[X]^2_{aq} = 6.4 \times 10^{-3} \text{ mol}^2 \text{ dm}^{-6} = 64 \times 10^{-4} \text{ mol}^2 \text{ dm}^{-6}$

$[X]_{aq} = 8.0 \times 10^{-2} \text{ mol dm}^{-3}$ **(4+1)**

$$K_D = \frac{[X]_{org-2}}{[X]_{aq}} = \frac{\left(\frac{0.2\ mol}{0.1\ dm^3}\ 0.08\ mol\ dm^{-3}\right)}{0.08\ mol\ dm^{-3}}$$ **(4+1)**

$K_D = 24$ **(4+1)**

Alternate answer for 6(c)

$K_D = \frac{(\frac{0.2\ mol - x}{0.1\ dm^3})}{(\frac{x}{0.1\ dm^3})}$ **(4+1)**

$x = \frac{0.2\ mol}{K_D + 1}$

$[X]_{aq} = \frac{\frac{0.2\ mol}{(K_D+1)}}{0.1\ dm^3} = \frac{2}{(K_D+1)} mol\ dm^{-3}$ **(4+1)**

$Rate\ =\ k\ [X]_{aq}^{m}\ [Y]_{aq}^{n}$

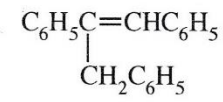
$6.4 \times 10^{-7} mol\ dm^{-3}\ s^{-1} = 1\ \times\ 10^{-3}\ mol\ dm^{-3}\ s^{-1}\ \left(\frac{2\ mol\ dm^{-3}}{(K_D+1)}\right)^2 (0.1\ mol\ dm^{-3})$ **(4+1)**

$64 \times 10^{-4} = \left(\frac{2}{K_D+1}\right)^2$ **(4+1)**

$K_D = 24$ **(4+1)**

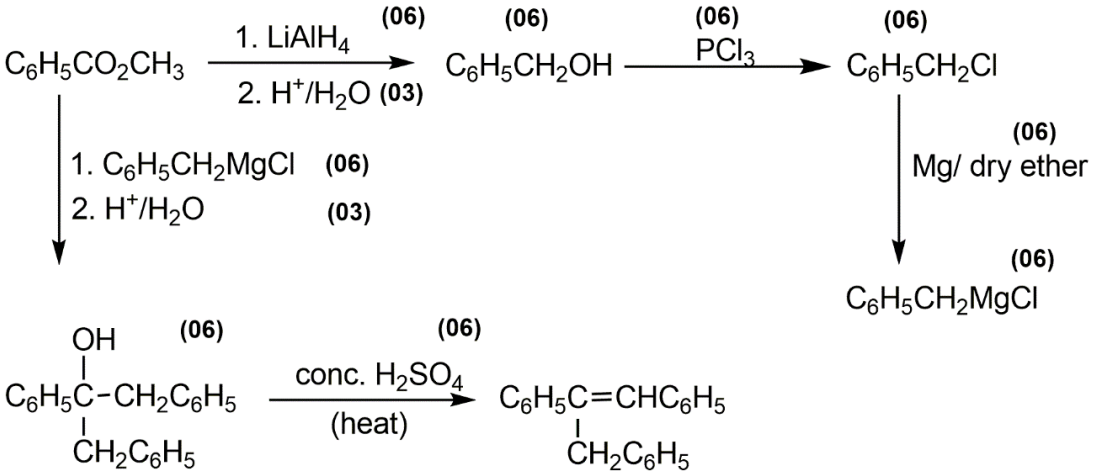
6 (c): 25 marks

7. (a) Using $C_6H_5CO_2CH_3$ as the only organic starting material and as reagents only those given in the list, show how you would synthesize the following compound in **not** more than **seven** (7) steps.



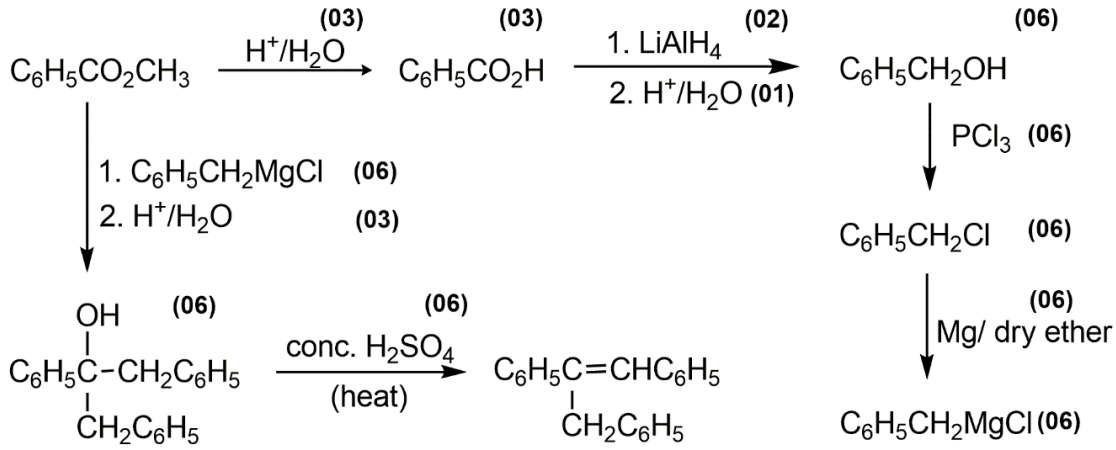
List of reagents

PCl_3, Mg/dry ether, H^+/H_2O, $LiAlH_4$, conc. H_2SO_4



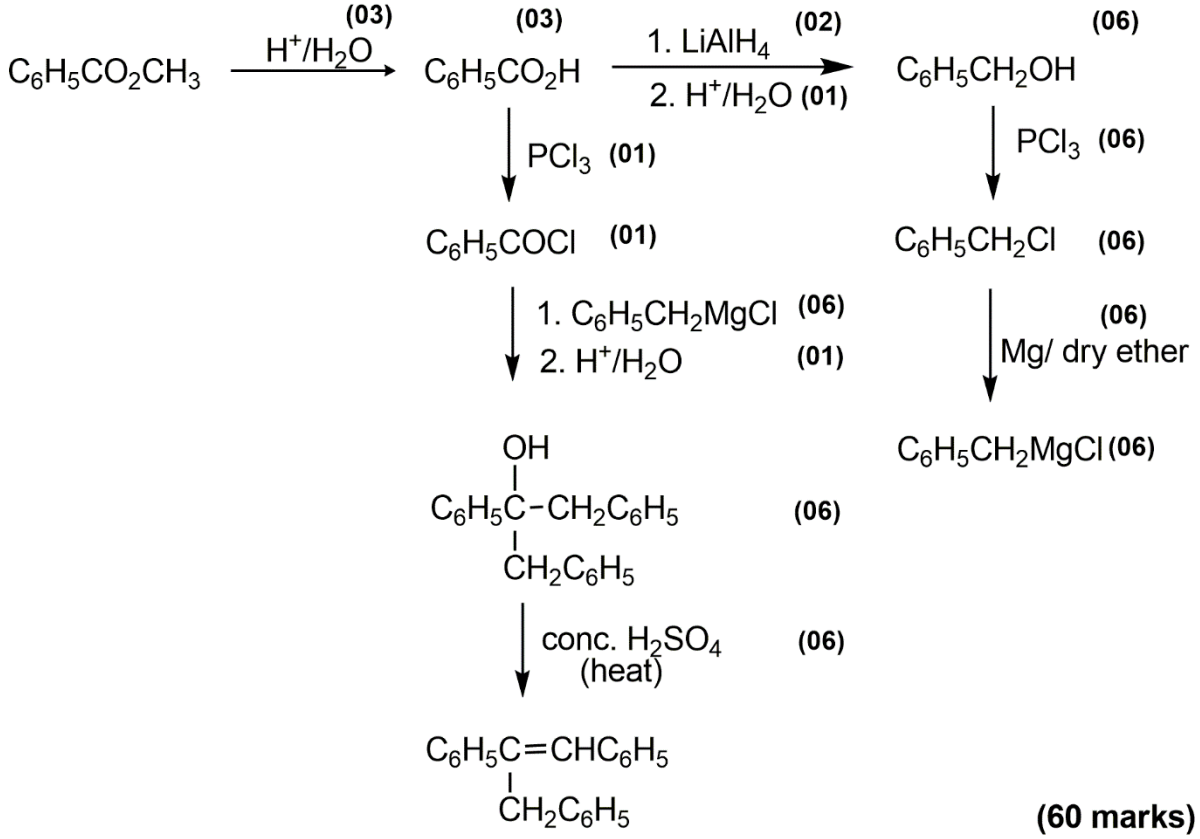
(60 marks)

Alternative Answer (I) for 7(a)



(60 marks)

Alternative Answer (II) for 7(a)



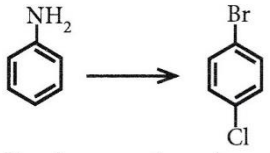
(60 marks)

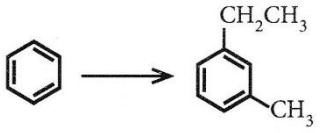
Note : 1. Do not award any marks if there are more than 7 steps.

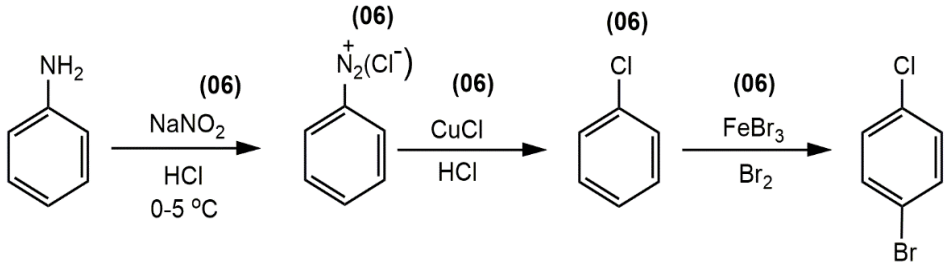
2. Do not count the hydrolysis steps subsequent to the reaction with Grignard reagent and with $LiAlH_4$ as separate steps.

7 (a) : 60 marks

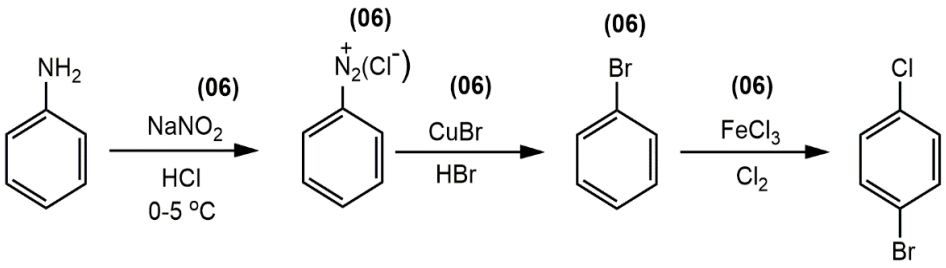
(*b*) Show how you would carry out each of the following conversions in **not** more than **three** (3) steps.

(i) 

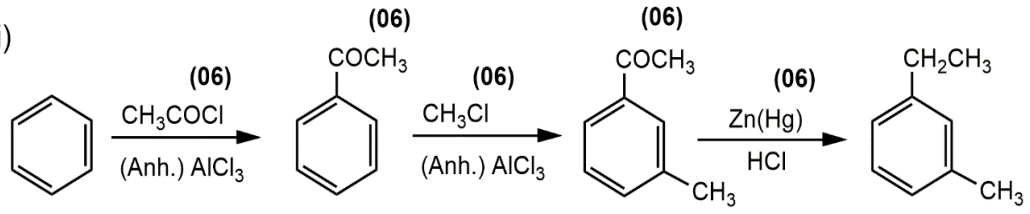
(ii) 

(b) (i) 

Alternative Answer for 7(b) (i)

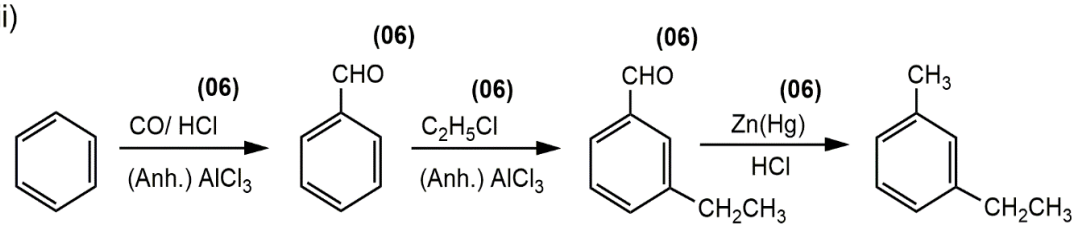


(30 marks)

(ii) 

(30 marks)

Alternative Answer for 7(b) (ii)

(ii) 

(30 marks)

7 (b) : 60 marks

(*c*) The following reaction gives two products.

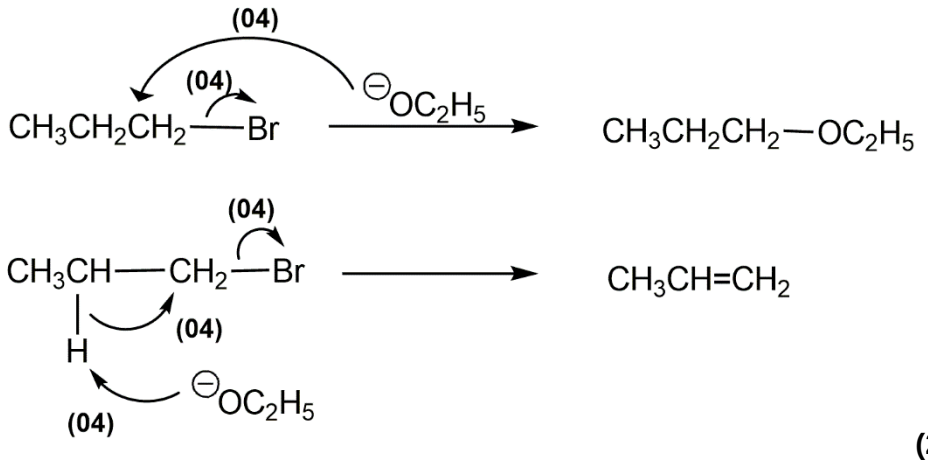
$$CH_3CH_2CH_2Br \xrightarrow{C_2H_5O^-}$$

(i) Write the structures of the two products.

(ii) Write the mechanisms for the formation of these two products.

(i). Products $CH_3CH_2CH_2\text{-}OC_2H_5$ $CH_3CH{=}CH_2$ **(05 + 05)**

(ii)



(20 marks)

7 (c) :30 marks

PART C — ESSAY

*Answer **two** questions only. (Each question carries **15 marks**.)*

8.(a) Solution **X** contains **four** metal cations. The following tests were carried out to identify these cations.

	Test	Observation
①	Dilute HCl was added to a small portion of **X**.	No precipitate.
②	H_2S was bubbled through the solution from ① above.	A black precipitate ($\mathbf{P_1}$)
③	$\mathbf{P_1}$ was separated by filtration. The filtrate was boiled to remove the H_2S, cooled, and NH_4Cl/NH_4OH was added.	A green precipitate ($\mathbf{P_2}$)
④	$\mathbf{P_2}$ was separated by filtration and H_2S was bubbled through the filtrate.	A white precipitate ($\mathbf{P_3}$)
⑤	$\mathbf{P_3}$ was separated by filtration. The filtrate was boiled to remove the H_2S, cooled, and $(NH_4)_2CO_3$ was added.	A white precipitate ($\mathbf{P_4}$)

The following tests were carried out on precipitates $\mathbf{P_1}$, $\mathbf{P_2}$, $\mathbf{P_3}$ and $\mathbf{P_4}$.

Precipitate	Test	Observation
$\mathbf{P_1}$	$\mathbf{P_1}$ was dissolved in hot dil. HNO_3 and conc. NH_4OH was added in excess.	A deep blue coloured solution (**solution 1**)
$\mathbf{P_2}$	* Excess dil. NaOH was added to $\mathbf{P_2}$ followed by H_2O_2. * Dilute H_2SO_4 was added to **solution 2**.	A yellow coloured solution (**solution 2**) An orange coloured solution (**solution 3**)
$\mathbf{P_3}$	* $\mathbf{P_3}$ was dissolved in dil. HCl and dil. NaOH was added gradually. * Addition of dil. NaOH was continued.	A white precipitate ($\mathbf{P_5}$) $\mathbf{P_5}$ dissolved to give a colourless solution (**solution 4**)
$\mathbf{P_4}$	$\mathbf{P_4}$ was dissolved in conc. HCl and subjected to the flame test.	A brick-red flame

(i) Identify the **four** metal cations in solution **X** (**Reasons are not required.**)

Cu^{2+}, Cr^{3+}, Zn^{2+}, Ca^{2+} **(05 marks x 4 = 20)**

(ii) Identify the precipitates $\mathbf{P_1}$, $\mathbf{P_2}$, $\mathbf{P_3}$, $\mathbf{P_4}$ and $\mathbf{P_5}$ and the chemical species responsible for the colours of **solutions 1, 2, 3** and **4**.

(**Note**: Write chemical formulae only.)

P_1: CuS

P_2: $Cr(OH)_3$

P_3: ZnS

P_4: $CaCO_3$

P_5: $Zn(OH)_2$ **(06 marks x 5 = 30)**

solution 1: $[Cu(NH_3)_4]^{2+}$ **(07)**

solution 2: Na_2CrO_4 **OR** CrO_4^{2-} **(06)**

solution 3: $Na_2Cr_2O_7$ **OR** $Cr_2O_7^{2-}$ **(06)**

solution 4: Na_2ZnO_2 **OR** ZnO_2^{2-} **OR** $Na_2Zn(OH)_4$ **OR** $[Zn(OH)_4]^{2-}$ **(06)**

8(a): 75 marks

(*b*) The water sample **Y** contains the anions SO_3^{2-}, SO_4^{2-} and NO_3^-. The following procedures were carried out for the quantitative analysis of the anions present in the water sample.

Procedure 1

To 25.00 cm^3 of sample **Y**, an excess of a dilute solution of $BaCl_2$ was added with stirring. Thereafter, excess dilute HCl was added with stirring to the precipitate formed until there was no further evolution of a gas with pungent odour. The solution was allowed to stand for 10 minutes and filtered. The precipitate was washed with distilled water and dried in an oven at 105 °C until a constant mass was obtained. The mass of the precipitate was 0.174 g. The filtrate obtained was kept for further analysis (**see procedure 3**).

Procedure 2

To 25.00 cm^3 of sample **Y**, an excess of dilute H_2SO_4 and acidified 5% KIO_3 solutions were added. The liberated I_2 was immediately titrated with 0.020 mol dm^{-3} $Na_2S_2O_3$ solution using starch as the indicator. The volume of $Na_2S_2O_3$ used was 20.00 cm^3. (Assume that in this procedure, SO_3^{2-} ions are oxidized to sulphate ions $\left(SO_4^{2-}\right)$ without any loss to the atmosphere.)

Procedure 3

The filtrate from **procedure 1** was neutralized with dilute NaOH and to it excess Al powder and dilute NaOH were added. The solution was heated and the gas evolved was transferred quantitatively to react with a 20.00 cm^3 volume of 0.11 mol dm^{-3} HCl solution. Completion of the reaction was tested with litmus. The HCl remaining after reacting with the gas evolved was titrated with 0.10 mol dm^{-3} NaOH solution using methyl orange as the indicator. The volume of NaOH required was 10.00 cm^3.

(i) Write balanced ionic/non-ionic equations for the reactions taking place in **procedures 1, 2 and 3**.

Procedure I

$SO_3^{2-} + BaCl_2 \rightarrow BaSO_3\downarrow + 2Cl^-$ **OR** $Ba^{2+} + SO_3^{2-} \rightarrow BaSO_3\downarrow$ **(02)**

$SO_4^{2-} + BaCl_2 \rightarrow BaSO_4\downarrow + 2Cl^-$ **OR** $Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4\downarrow$ **(02)**

$BaSO_3\downarrow + 2HCl \rightarrow BaCl_2 + SO_2 + H_2O$ **(03)**

$BaSO_4$ will remain insoluble

Procedure 2

$2IO_3^- + 12H^+ + 10e \rightarrow I_2 + 6H_2O$ **(02)**

$5(SO_3^{2-} + H_2O \rightarrow SO_4^{2-} + 2H^+ + 2e)$ **(02)**

$2IO_3^- + 5SO_3^{2-} + 2H^+ \rightarrow I_2 + 5SO_4^{2-} + H_2O$ **(03)**

OR

$2IO_3^- + 12H^+ + 10e \rightarrow I_2 + 6H_2O$ **(02)**

$5(SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e)$ **(02)**

$2IO_3^- + 5SO_2 + 4H_2O \rightarrow I_2 + 5SO_4^{2-} + 8H^+$ **(03)**

$I_2 + 2e \rightarrow 2I^-$ **(02)**

$2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2e$ **(02)**

$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$ **(03)**

Therefore, $5SO_3^{2-} \equiv 2S_2O_3^{2-}$ **OR** $5SO_2 \equiv 2S_2O_3^{2-}$ & $SO_2 \equiv SO_3^{2-}$ **(02)**

Procedure 3

$3NO_3^- + 8Al + 5OH^- + 2H_2O \rightarrow 8AlO_2^- + 3NH_3$ **(02)**

$NH_3 + HCl \rightarrow NH_4Cl$ **(02)**

$HCl + NaOH \rightarrow NaCl + H_2O$ **(02)**

(ii) Determine the concentrations (mol dm^{-3}) of SO_3^{2-}, SO_4^{2-} and NO_3^- in water sample **Y**. (Ba = 137; S = 32; O = 16)

Procedure I – Determination of SO_4^{2-}

Molar mass of $BaSO_4$ = 137 + 32 + 64 = 233 **(02)**

Mass of $BaSO_4$ precipitate = 0.174 g

Therefore, moles of $BaSO_4$ = $\frac{0.174}{233}$ **(02)**

Therefore, moles of SO_4^{2-} = $\frac{0.174}{233} = 7.47 \times 10^{-4}$ **(02)**

Concentration of SO_4^{2-} = $\frac{7.47 \times 10^{-4}}{25} \times 1000$ **(02)**

= **0.029 (0.03) mol dm^{-3}** **(03 + 01)**

Procedure 2 – Determination of SO_3^{2-}

Moles of $S_2O_3^{2-}$ = $\frac{0.02}{1000} \times 20$ **(02)**

Therefore, moles of SO_3^{2-} = $\frac{0.02}{1000} \times 20 \times \frac{5}{2}$ **(02)**

Concentration of SO_3^{2-} = $\frac{0.02}{1000} \times 20 \times \frac{5}{2} \times \frac{1000}{25}$ **(02)**

= **0.04 mol dm^{-3}** **(03 + 01)**

Procedure 3 – Determination of NO_3^-

Moles of HCl = $\frac{0.11}{1000} \times 20$ **(02)**

Moles of NaOH = $\frac{0.10}{1000} \times 10$ **(02)**

Since NaOH and HCl react in the ratio of 1:1

Moles of HCl reacted with NH_3 = $\frac{0.11}{1000} \times 20 - \frac{0.10}{1000} \times 10$ **(02)**

= $\frac{1}{1000}(2.2 - 1) = \frac{1.2}{1000}$ **(02)**

Therefore, moles of NH_3 = $\frac{1.2}{1000}$ **(02)**

Therefore, moles of NO_3^- = $\frac{1.2}{1000}$ **(02)**

Concentration of NO_3^- = $\frac{1.2}{1000} \times \frac{1000}{25}$ **(02)**

= **0.048 mol dm^{-3}** **(03 + 01)**

(iii) Give colour changes that would be observed in the titrations in **procedures 2** and **3**.

(**Note**: Assume that other ions that may interfere with the analysis are not present in sample **Y**.)

Procedure 2: Blue → Colourless **(03)**

Procedure 3: Red → Yellow **(03)**

8(b): 75 marks

9. (*a*) The following questions are based on the Solvay process used in the manufacture of sodium carbonate.

(i) State the raw materials used in the Solvay process.

Brine / concentrated NaCl solution **(03)**

Lime stone / calcite/ $CaCO_3$ **(03)**

NH_3 **(03)**

(ii) Briefly explain the manufacturing process of sodium carbonate, with the aid of balanced chemical equations where applicable.

CO_2 is obtained by heating $CaCO_3$ **(02)**

$CaCO_3 \rightarrow CaO + CO_2$ **(03)**

NH_3 is dissolved in brine solution. **(02)**

$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$ ---------(A) **(03)**

Using the counter current principle **(03)**

CO_2 is dissolved in ammoniacal brine solution. **(02)**

$OH^- + CO_2 \rightarrow HCO_3^-$ ----------(B) **(03)**

When HCO_3^- concentration increases, $NaHCO_3$ precipitates. **(02)**

$Na^+ + HCO_3^- \rightarrow NaHCO_3$ ----------(C) **(03)**

OR

Reactions (A), (B) and (C) can be combined.

$NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$ **(09)**

For three descriptions above **(02 x 3)**

OR

Reactions (A), (B), (C) and (D) can be combined.

$2NaCl + 2NH_3 + CO_2 + H_2O \rightarrow Na_2CO_3 + 2NH_4Cl$ **(12)**

For four descriptions above **(02 x 4)**

Low temperature is maintained to increase the solubility of gases. **(02)**

Low temperature reduces the solubility of $NaHCO_3$ **(01)**

Heating of $NaHCO_3$ to produce Na_2CO_3 **(02)**

$2NaHCO_3 \rightarrow Na_2CO_3. + CO_2 + H_2O$ ----------(D) **(03)**

Recovery of NH_3 **(02)**

$Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + 2H_2O$ **(03)**

OR

$CaO + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + H_2O$ **(03)**

(iii) Give **three** uses of sodium carbonate.

- Softening of hard water
- Manufacture of soap
- Manufacture of glass
- Manufacture of detergents
- Manufacture of paper

(Any three) **(03 x 3 = 09 marks)**

(iv) One of the raw materials identified in (i) above, is used to manufacture two important chemical substances; one has disinfection properties whereas the other is used in the welding industry. Identify the **two** chemical substances and give balanced chemical equations to show their formation.

Raw material – $CaCO_3$ **(02)**

Disinfectant – bleaching powder **(02)**

Welding industry – CaC_2 **OR** calcium carbide **(02)**

Manufacture of bleaching powder

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ **(03)**

$CaO + H_2O \longrightarrow Ca(OH)_2$ **(03)**

$3Ca(OH)_2 + 2Cl_2(g) \longrightarrow Ca(OCl)_2.Ca(OH)_2.CaCl_2.2H_2O$ **(03)**
(bleaching powder)

Manufacture of CaC_2

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ **(03)**

$CaO + 3C \xrightarrow[2000\ ^oC]{\Delta} CaC_2 + CO$ **(02)** **(01)**

OR

$2CaO + 5C \xrightarrow[2000\ ^oC]{\Delta} 2CaC_2 + CO_2$ **(02)** **(01)**

Note: Heating is not required for award of marks

9(a): 75 marks

(*b*) The following questions are based on the earth and its environment.

(i) The atmosphere has been divided into several regions based on the temperature variation

I. Name the **two** regions that are closest to the Earth.

II. In which of these is the ozone layer located?

I. Troposphere and stratosphere **(04 + 04)**

II. Stratosphere **(04)**

(ii) State **two** major nitrogen species present in each of the following spheres.

I. Atmosphere

II. Hydrosphere

Atmosphere – N_2 and **either** NO_2 / NO / NOx **(03 + 03)**

Hydrosphere – NO_2^-, NO_3^- , NH_4^+ **(Any two)** **(03 + 03)**

(iii) State **two** major carbon species present in each of the following spheres.

I. Atmosphere

II. Hydrosphere

Atmosphere – CO_2 and **either** CO / CH_4 **(03 + 03)**

Hydrosphere – CO_3^{2-}, HCO_3^-, $CO_2(aq)$, H_2CO_3
(Any two) **(03 + 03)**

(iv) State **two** important processes in each of the following cycles.

I. Carbon cycle

II. Hydrological cycle

Carbon cycle – photosynthesis, combustion, respiration, decay, sedimentation, dissolution
(Any two) **(03 + 03)**

Hydrological cycle – evaporation, precipitation, ground water/ river water discharge to the oceans
(Any two) **(03 + 03)**

Note : Consider only the first two answers for 10 (b) (ii) – (iv)

(v) Identify the nitrogen fixing processes associated with the nitrogen cycle giving balanced chemical equations where applicable.

Atmospheric fixation – $N_2 + O_2 \rightarrow 2NO$ **(02 + 02)**

Biological fixation – ----- **(02)**

Industrial fixation – $N_2 + 3H_2 \rightarrow 2NH_3$ **(02 + 02)**

(vi) Identify **two** nitrogen containing organic compounds that are produced during the formation of photochemical smog.

peroxyacetylnitrate **OR** PAN **(04)**

peroxybenzoylnitrate **OR** PBN **(04)**

Note : Consider only the first two answers

(vii) Earth is subjected to numerous adverse effects due to acid rain. List **three** of them.

- Damage to plants
- Death of fish in lakes
- Metallic structures, bridges, motor vehicles are affected
- Hardness of water increases
- Acidity of water increases
- Salinity of water increases
- Heavy metal levels of water increases
- Decay of calcium carbonate based structures
- Dissolution of mineral deposits

(Any three) **(03 + 03 + 03)**

9(b): 75 marks

10. *(a)* **A**, **B**, **C** and **D** are coordination compounds. They have an octahedral geometry. In each compound **two types** of ligands are coordinated to the metal ion. The molecular formulae of the compounds are (**not** in order):

$NiCl_2H_{12}N_4$, $NiI_2H_{16}N_4O_2$, $NiH_{16}N_6O_8$ and $NiCl_2H_{15}N_3O_3$

The observations when aqueous solutions of the compounds are treated with $Pb(CH_3COO)_2(aq)$ are given below.

Compound	$Pb(CH_3COO)_2(aq)$
A	A white precipitate that is soluble in hot water
B	No precipitate
C	A yellow precipitate that is soluble in hot water
D	No precipitate

(i) Give the structures of **A**, **B**, **C** and **D**. All the ligands in compound **B** are coordinated to the metal ion.

A: $[Ni(H_2O)_3(NH_3)_3]Cl_2$ **OR** $[Ni(NH_3)_3(H_2O)_3]Cl_2$ **(06)**

B: $[Ni(NH_3)_4Cl_2]$ **OR** $[NiCl_2(NH_3)_4]$ **(06)**

C: $[Ni(NH_3)_4(H_2O)_2]I_2$ **OR** $[Ni(H_2O)_2(NH_3)_4]I_2$ **(06)**

D: $[Ni(NH_3)_4(H_2O)_2](NO_3)_2$ **OR** $[Ni(H_2O)_2(NH_3)_4](NO_3)_2$ **(09)**

Note: OH_2 may be used instead of H_2O.

(ii) Write the chemical formulae of the precipitates formed on treatment of the compounds with $Pb(CH_3COO)_2(aq)$.
(**Note**: Indicate compound and reagent.)

A with $Pb(CH_3COO)_2$ $PbCl_2\downarrow$ **(03)**

C with $Pb(CH_3COO)_2$ $PbI_2\downarrow$ **(03)**

(iii) Give the IUPAC names of **A**, **B**, **C** and **D**.

A: triamminetriaquanickel(II) chloride **(06)**

B: tetraamminedichloridonickel(II) **(06)**

C: tetraamminediaquanickel(II) iodide **(06)**

D: tetraamminediaquanickel(II) nitrate **(06)**

(iv) State a chemical test, together with the observation, to identify each of the anion/s if present, that is/are not coordinated to the metal ion in the compounds given above.
(**Note**: The tests given by you should not be a test stated here.)

Cl^- Add a solution of $AgNO_3$. **(03)**

A white precipitate is formed. The white precipitate dissolves in dilute NH_4OH. **(03)**

I^- Add a solution of $AgNO_3$. **(03)**

A yellow precipitate is formed. The yellow precipitate does not dissolve in conc. NH_4OH. **(03)**

OR

Add a few drops of $CHCl_3$ and then a little Cl_2 water. **(03)**
Shake the tube.
$CHCl_3$ layer turns violet. **(03)**

NO_3^- Add a freshly prepared solution of $FeSO_4$. Then add a few drops of conc. H_2SO_4 slowly down the side of the test-tube. **(03)**
A brown ring is seen at the interface of the two solutions. **(03)**

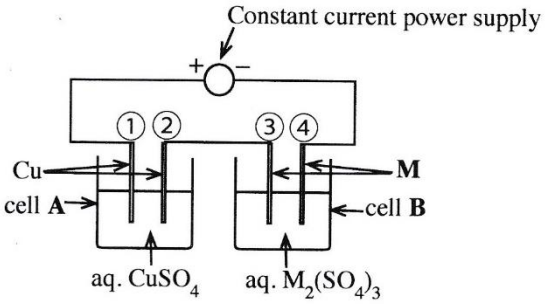
OR

Add Al powder and NaOH(aq) to the solution and boil. **(03)**
Smell of NH_3/ Gas that turns Nestler reagent brown is evolved. **(03)**

10(a): 75 marks

(b) The setup shown in the figure was used to find the relative atomic mass of the metal, **M**.

The electrolysis was carried out for 10 minutes using a constant current. The mass of the cathode in cell **A** was increased by 31.75 mg whereas the mass of the cathode in cell **B** increased by 147.60 mg during this time period. (Assume that the electrolysis of water does not take place in cells **A** and **B**.)



(i) Identify the anode and cathode in each of the cells **A** and **B** (in terms of the numbers ①, ②, ③, and ④).

Cell A

Anode = 1 **(5)**

Cathode = 2 **(5)**

Cell B

Anode = 3 **(5)**

Cathode = 4 **(5)**

(ii) Write the half reaction taking place at each electrode in each cell.

Electrode reactions

Cell A electrode 1	$Cu(s) \rightarrow Cu^{2+}(aq) + 2e$	**(6)**
Cell A electrode 2	$Cu^{2+}(aq) + 2e \rightarrow Cu(s)$	**(6)**
Cell B electrode 3	$M(s) \rightarrow M^{3+}(aq) + 3e$	**(6)**
Cell B electrode 4	$M^{3+}(aq) + 3e \rightarrow M(s)$	**(6)**

Note : physical states must be given

(iii) Calculate the constant current used in electrolysis.

Amount of Cu(s) dissolved = 31.75×10^{-3} g

Charge required for this = $\frac{2 \times 96500 \text{ c mol}^{-1} \times 31.75 \times 10^{-3}\text{g}}{63.5 \text{ g mol}^{-1}}$ = i x 10 x 60 s **(1+1)+(1+1)+(1+1)+(1+1)**

Correct stoichiometry **(5)**

The current used in the electrolysis = i = 0.16 A **(4+1)**

Alternative Answer for 10(b) (iii)

Amount of Cu deposited = $\frac{31.75 \times 10^{-3}\ g}{63.5\ g\ mol^{-1}}$ **(1+1)**

= 0.5×10^{-3} mol

Amount of charge required = $0.5 \times 10^{-3} \times 2$ mol = 10^{-3} mol For using correct stoichiometry **(5)**

= 10^{-3} mol × 96500 C mol^{-1} **(1+1)**

= 96.5 C

Current = $\frac{96.5\ C}{10 \times 60\ s}$ **(1+1)**

= 0.16 A **(4+1)**

(iv) Calculate the relative atomic mass of metal, **M**.

Increase in mass of the electrode 4 in cell B is due to deposition of M(s)
Increase in mass = 147.6x 10^{-3} g
Amount of M deposited = 147.6 x 10^{-3} g /W
W=molecular weight of M
Charge needed for this = $\frac{3 \times 96500 \text{ c mol}^{-1} \times 147.6 \times 10^{-3} \text{ g}}{W}$ = 0.16 A x 600 s **(1+1)+(1+1)+(1+1)**

Correct stoichiometry **(5)**

W = 445.1 g mol^{-1} **(1+1)**

Alternative Answer (I) for 10(b) (iv)

The amount of charge flown is equal.
M mol x 3 = Cu mol x 2

$$\frac{147.6 \times 10^{-3}\ g \times 3\ mol}{W} = \frac{31.75 \times 10^{-3}\ g \times 2\ mol}{63.5\ g\ mol^{-1}}$$ **For using correct stoichiometry (5)**

W = $\frac{147.6 \times 3 \times 63.5}{31.75 \times 2}$ g mol^{-1}

= 442.8 g mol^{-1} **(1+1)**

Alternative Answer (II) for 10(b) (iv)

Amount of M deposited = Amount of charge flown / 3
= $\frac{10^{-3}}{3}\ mol$ **For using correct stoichiometry (5)**

Molar mass of M = $\frac{147.6 \times 10^{-3}\ g}{\frac{10^{-3}}{3}\ mol}$
= 147.6 × 3 g mol^{-1}
= 442.8 g mol^{-1} **(1+1)**

Note : If symbols (or any other values) are used for the atomic mass of Cu and Faraday constant and the answers are provided with those symbols or using those values, award full marks.

10 (b): 75 marks

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